# Trace-Tempered Choice

Lean-faithful appendices to Theorem 1

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**Standing notation.** All payoff, action, and record alphabets are finite, and action and record alphabets are nonempty. The primitive relations, common-payoff block environments, and Axioms (A1)–(A8) are those in the paper. A payoff-preserving record processor $T$ and an action report $S$ satisfy, respectively,

$$(KT)(o, r' \mid a) = \sum_r K(o, r \mid a) T(r' \mid o, r), \tag{1}$$

$$(qS)(b)\widehat{K}(o, r \mid b) = \sum_a q(a) S(b \mid a) K(o, r \mid a). \tag{2}$$

The second identity is imposed without division and hence leaves rows at null reported actions unrestricted.

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## A Lean-faithful proof of Theorem 1

The proof has one substantial auxiliary input: the pure-trace proposition stated next and proved from the primitive pure-record conditions in Appendix B. Everything after that input is the calibrated assembly used by the checked Lean proof. We also use the ordinary mixture-space theorem: a complete and transitive mixture order that satisfies independence and admits calibration on every closed mixture segment has an affine representation.

The logical order is therefore

$$
\begin{tikzcd}[column sep=tiny]
& \boxed{\text{Axioms (A1)–(A8)}} \arrow[dl] \arrow[dr] & \\
\boxed{\begin{array}{c}\text{terminal-law quotient and}\\ \text{affine material scale}\end{array}} \arrow[dr] & & \boxed{\begin{array}{c}\text{pure-trace proposition (Appendix B) and}\\ \text{one common trace scale}\end{array}} \arrow[dl] \\
& \boxed{\text{branch calibration and payoff telescope}} \arrow[d, Rightarrow] & \\
& \boxed{\text{sequentialisation and Theorem 1}}. &
\end{tikzcd}
$$

The left branch identifies the value of terminal payoffs and fixes a common material scale. The right branch is proved once on the constant-payoff domain. They meet in the deterministic branch calculation, which aligns material increments with the trace scale and telescopes across all reached branches.

## A.1 The auxiliary pure-trace proposition

Appendix B defines the pure-record domain, the block comparison notation, Conditions (P1)–(P5), and the mutual information $I_{q,P}(A;X)$. The exact input is stated here for use before it is proved; Appendix B states it formally as Proposition 9. Thus there is one auxiliary proposition, not two separately numbered copies, and Theorem 1 remains the unique theorem of the joint payoff–record model.

> **Pure-trace proposition.** For a family $\{\succeq_P\}_P$ on all finite record channels described in Appendix B, the following are equivalent.
>
> (i) The family satisfies Conditions (P1)–(P5).
>
> (ii) For every pair of finite sets $A, X$, every channel $P : A \to \Delta(X)$, and all $q, q' \in \Delta(A)$,
> $$q \succeq_P q' \quad\Longleftrightarrow\quad I_{q,P}(A;X) \geq I_{q',P}(A;X).$$
>
> Moreover, on every finite block environment $\bigsqcup_{k\in K} P_k$, all block-supported cross-channel comparisons use that same information scale: for distinct $i, j \in K$ and all $q_i \in \Delta(A_i)$, $q_j \in \Delta(A_j)$,
> $$q_i^i \succeq_{\bigsqcup_{k\in K} P_k} q_j^j \quad\Longleftrightarrow\quad I_{q_i,P_i}(A_i;X_i) \geq I_{q_j,P_j}(A_j;X_j).$$

Appendix B proves both directions and the moreover clause. Appendix A uses only (i)⇒(ii) and its common-scale clause, applied to the constant-payoff lift.

## A.2 Preliminaries

We first dispose of three pieces of bookkeeping. They will be used without further comment.

Recall that $(q, K) \succeq (p, L)$ denotes the paper's canonical two-block comparison. For channels $E, G$ at a common prior $q$, write $E \succeq_q G$ for $(q, E) \succeq (q, G)$. A *finite experiment at* $q$ is a finite payoff–record channel paired with that prior; its marked terminal law is denoted by $\eta_E$.

**Lemma 1** (Blocks, supports, and dummies)**.** *Suppose Axioms* (A1)–(A5) *hold.*

(a) *After identifying alternatives related by invertible action or record relabellings, the derived comparison of pairs is complete and transitive. If either member of a pair comparison is replaced by an indifferent alternative, the comparison is unchanged.*

(b) *If $S = \operatorname{supp} q$, then $(q, K)$ is indifferent to the restriction of $(q, K)$ to $S$.*

(c) *If $s \in \Delta(B)$ and $K^{\uparrow}(o, r \mid a, b) := K(o, r \mid a)$, then*

$$(q \otimes s, K^{\uparrow}) \sim (q, K). \tag{3}$$

*Proof.* Place any finite collection of alternatives in one finite block environment. Axiom (A3) removes unused blocks, and Axiom (A1) supplies completeness and transitivity there. Invertible action and record relabellings are neutral by applying Axioms (A4) and (A5) in both directions. In particular, swapping both tags identifies the two orientations of a two-block comparison. This proves part (a); a three-block environment gives transitivity and a four-block environment gives replacement.

For part (b), report an action in $S$ by its inclusion in $A$. In the other direction, report $a \in S$ when the realised action is $a$, and choose an arbitrary element of $S$ off the support. Both reports satisfy (2); rows attached to null reported actions may be completed arbitrarily. Axiom (A5) gives the two comparisons.

For part (c), first project $(a, b)$ to $a$. Conversely, given $a$, draw $b$ independently from $s$ and report $(a, b)$. Again the two joint-law identities in (2) are exact, so Axiom (A5) applies in both directions. Part (a) completes both arguments. □

Fix for the moment a full-support prior $q \in \Delta(A)$. For a joint channel $K$, write

$$p_K(o, r) := \sum_a q(a) K(o, r \mid a), \qquad \pi^K_{or}(a) := \frac{q(a)K(o, r \mid a)}{p_K(o, r)}$$

when $p_K(o, r) > 0$, and define its *marked terminal law*

$$\eta_K := \sum_{(o,r): p_K(o,r)>0} p_K(o, r)\, \delta_{(o, \pi^K_{or})}. \tag{4}$$

It records the terminal payoff and the posterior about the realised action. Write $I_q(E)$ for the mutual information between the action and the complete visible output of $E$ under $q$. When the output is a first-stage record $Y$, we also write this quantity as $I_q(A; Y)$.

**Lemma 2** (Terminal laws and affine fibres)**.** *Under Axioms* (A1)–(A6)*, the following statements hold.*

(a) *If $q$ has full support and $\eta_K = \eta_L$, then $(q, K) \sim (q, L)$.*

(b) *At every full-support $q$, the quotient of finite experiments by their marked terminal laws has an affine representation $W_q$.*

*Proof.* We give the finite Blackwell argument behind part (a). For a payoff $o$ and posterior $\pi$, let

$$M_K(o, \pi) := \sum_{\substack{r:\, p_K(o,r)>0 \\ \pi^K_{or}=\pi}} p_K(o, r).$$

Equality of marked laws gives $M_K(o, \pi) = M_L(o, \pi) =: M(o, \pi)$. For $p_K(o, r) > 0$, draw a record $s$ of $L$ according to

$$T(s \mid o, r) := \mathbf{1}_{\{p_L(o,s)>0,\, \pi^L_{os}=\pi^K_{or}\}} \frac{p_L(o, s)}{M(o, \pi^K_{or})}. \tag{5}$$

Choose an arbitrary probability row when $p_K(o, r) = 0$. This is a payoff-preserving record processor. Bayes' formula gives

$$q(a)K(o, r \mid a) = p_K(o, r)\pi^K_{or}(a).$$

For a reached target record $s$, with posterior $\pi = \pi^L_{os}$, it follows that

$$\sum_r K(o, r \mid a) T(s \mid o, r) = \frac{p_L(o, s)}{M(o, \pi)} \frac{\pi(a)}{q(a)} M(o, \pi) = L(o, s \mid a).$$

If $s$ is unreached, full support makes both sides zero. Thus postprocessing $K$ by $T$ gives $L$. The symmetric construction gives a processor from $L$ to $K$. Axiom (A4) and Lemma 1(a) give indifference. For a boundary prior, Lemma 1(b) first deletes the null actions.

For part (b), take the quotient of actual finite marked experiments by equality of (4). If $E$ and $G$ are two such experiments, their public mixture first draws a coin independent of the action, runs the selected experiment, and retains the coin in the record. Its marked law is

$$\eta_{tE \oplus (1-t)G} = t\eta_E + (1-t)\eta_G. \tag{6}$$

Axiom (A6) gives, for $0 < t < 1$ and any third experiment $L$,

$$E \succeq_q L \quad \Longleftrightarrow \quad tE \oplus (1-t)G \succeq_q tL \oplus (1-t)G. \tag{7}$$

For the reverse implication, if $E \not\succeq_q L$, completeness gives $L \succ_q E$; the strict part of Axiom (A6) makes the reverse public mixture strict. If record alphabets differ, put them in a common disjoint union before invoking the axiom; part (a) identifies the result with (6).

It remains only to check the Archimedean step. A closed segment between two experiments, and any fixed comparator, can be realised on one fixed union of their finite record alphabets. Its channel entries vary continuously with $t \in [0, 1]$. Axiom (A2) therefore makes both contour sets along the segment closed. Completeness says that they cover $[0, 1]$; when a target lies between the two endpoints, the sets contain opposite endpoints and connectedness makes them intersect. Thus every such target is indifferent to a point of the segment. The mixture-space theorem now supplies an affine representative $W_q$. This quotient is closed under every mixture and embedding used below. □

### A.3 The two common scales

**Lemma 3** (Material utility and a common affine scale)**.** *Under Axioms* (A1)–(A7)*, there are outcomes $o^+, o^-$ and a nonconstant function $u : O \to \mathbb{R}$ such that*

$$u(o^+) = 1, \qquad u(o^-) = 0.$$

*The payoff-lottery order is represented by $\sum_o \ell(o)u(o)$. The representatives in Lemma 2 may be normalised so that*

$$W_q(\ell) = \sum_o \ell(o)u(o). \tag{8}$$

*With this normalisation, adjoining a full-support independent dummy to a full-support prior preserves value, and for full-support $q, q'$,*

$$(q, E) \succeq (q', G) \quad \Longleftrightarrow \quad W_q(E) \geq W_{q'}(G). \tag{9}$$

*Proof.* On a singleton action set, terminal-law sufficiency identifies an experiment with its payoff lottery. Public-mixture independence and continuity, proved as in Lemma 2, give an affine utility on $\Delta(O)$. Axiom (A7) supplies $o^+ \succ o^-$. Normalise their values to one and zero and let $u(o)$ be the value of the degenerate lottery at $o$. Affinity on the finite simplex gives expected utility.

An action report may ignore its input and draw an arbitrary target prior. Axiom (A5), applied in both directions, therefore makes this payoff lottery order independent of the prior and action alphabet. Normalise each $W_q$ at the same two deterministic payoffs. The payoff-lottery embedding is affine and order-reflecting, so positive-affine uniqueness gives (8).

Dummy lifting between full-support fibres is likewise an affine order embedding. Positive-affine uniqueness initially permits

$$W_{q\otimes s}(E^{\uparrow}) = cW_q(E) + d, \qquad c > 0.$$

The two payoff anchors have values zero and one on both sides; hence $c = 1$ and $d = 0$. To compare $E$ at $q$ with $G$ at $q'$, lift each to the common product prior $q \otimes q'$, using the other prior as the dummy. Equation (3), the replacement part of Lemma 1, and the common product representative give (9). □

We now use Proposition 9. Apply it to the constant-payoff lift

$$K_P(o, r \mid a) = \mathbf{1}_{\{o=o^-\}} P(r \mid a).$$

This lift commutes with block formation, record processing, action reporting, limits on each finite simplex, and compounding. Thus Axioms (A1) and (A8) give the pure weak order and its positive orientation; Axiom (A2) gives continuity; Axiom (A3) gives block coherence; Axioms (A4) and (A5) give the two data processing conditions, including every null-row completion; and Axiom (A6) gives branchwise continuation. These are precisely the hypotheses of Proposition 9. Its conclusion is that every within-channel and block-supported comparison between such lifts is ordered by $I_{q,P}(A; R)$.

**Lemma 4** (The trace scale)**.** *Under Axioms* (A1)–(A8)*, there is one number* $\lambda > 0$*, independent of the prior and all finite alphabets, such that every full-support, nontrivial action fibre and every experiment* $E$ *that pays* $o^-$ *surely satisfy*

$$W_q(E) = \lambda I_q(E). \tag{10}$$

*Proof.* Fix a nontrivial full-support $q$. On the constant-$o^-$ face, $W_q$ is affine by Lemma 2. For any experiment $E$, the finite entropy identity

$$I_q(E) = H(q) - \sum_{(o,\pi)\in \operatorname{supp} \eta_E} \eta_E(o, \pi) H(\pi). \tag{11}$$

shows that mutual information depends only on the marked terminal law and is affine under public mixtures. Proposition 9 says that $W_q$ and $I_q$ represent the same nonconstant order on the constant-low face. Positive-affine uniqueness therefore gives

$$W_q(K_P) = \lambda_q I_{q,P} + b_q, \qquad \lambda_q > 0.$$

The uninformative experiment has mutual information zero and, by (8), value $u(o^-) = 0$. Thus $b_q = 0$.

The coefficient does not depend on $q$. Let $q$ and $q'$ be nontrivial full-support priors. Lift full revelation at $q$ to $q \otimes q'$ by adding an independent dummy coordinate. Normalised value and mutual information are both unchanged. Since the common information value is $H(q) > 0$, cancellation gives $\lambda_{q\otimes q'} = \lambda_q$. Lifting full revelation from the other coordinate gives $\lambda_{q\otimes q'} = \lambda_{q'}$. Hence all such coefficients agree. Equivalently, fix the uniform prior on a two-point dummy alphabet and call its coefficient $\lambda$. Axiom (A8) fixes the strict orientation, so $\lambda > 0$.

Finally, any experiment that pays $o^-$ surely is a pure record experiment after the constant payoff coordinate is deleted; restoring that coordinate changes neither its marked terminal law nor its mutual information. This proves (10). □

## A.4 The branch calculation

Let $q$ be full support on a nontrivial $A$ and let $P : A \to \Delta(Y)$ be a first-stage record channel. Put

$$m_y := \sum_a q(a)P(y \mid a).$$

For a deterministic payoff profile $g : Y \to O$, let $C(P, g)$ denote the compound that first runs $P$ and then pays $g(y)$ with no further record. Write $g_y^o$ for the profile that pays $o$ at $y$ and $o^-$ elsewhere.

**Lemma 5** (The reached-branch payoff increment)**.** *For every $y \in Y$ and $o \in O$,*

$$W_q(C(P, g_y^o)) - W_q(C(P, o^-)) = m_y u(o). \tag{12}$$

*Proof.* If $m_y = 0$, changing the payoff on branch $y$ does not change the marked terminal law, so both sides are zero. Suppose $m_y > 0$, and put

$$r_y(a) := \frac{q(a)P(y \mid a)}{m_y}, \qquad S_y := \operatorname{supp} r_y.$$

Let $\bar r_y$ be the full-support restriction of $r_y$ to $S_y$. For a marked continuation $E$ on $S_y$, extend $E$ to $A$ through a support projection, insert it at branch $y$, and put the uninformative payoff $o^-$ at every other branch. Denote the resulting experiment by $J_y(E)$. If $a \notin S_y$, then $r_y(a) = 0$; since $q(a) > 0$ and $m_y > 0$, the definition of $r_y$ gives $P(y \mid a) = 0$. Hence the continuation rows chosen outside $S_y$ are never used on branch $y$. Support restriction identifies their local versions as well, so the extension is independent of the chosen projection.

Support restriction and Axiom (A6), including its strict part, give

$$E \succeq_{\bar r_y} G \iff J_y(E) \succeq_q J_y(G).$$

For the reverse implication, failure of the local weak comparison gives a strict reverse comparison by completeness, and strict branchwise consistency gives a strict reverse compound comparison. Different continuation record alphabets are first put in a common disjoint union. Moreover, $J_y$ commutes with public mixtures at the level of marked terminal laws. Thus affine uniqueness gives constants $c_y > 0, d_y$ such that

$$W_q(J_y(E)) = c_y W_{\bar r_y}(E) + d_y. \tag{13}$$

Suppose first that $S_y$ has at least two elements. At payoff $o^-$, compare full revelation, $E^{\mathrm{id}}$, with an uninformative record, $E^0$. Lemma 4 gives the local difference

$$W_{\bar r_y}(E^{\mathrm{id}}) - W_{\bar r_y}(E^0) = \lambda H(\bar r_y).$$

The finite mutual-information chain rule for insertion is

$$I_q(J_y(E)) = I_q(A; Y) + m_y I_{\bar r_y}(E). \tag{14}$$

Support extension does not change the last term. Both inserted experiments pay $o^-$ surely, so Lemma 4 and (14) make their aggregate value difference $\lambda m_y H(\bar r_y)$. Equation (13) makes the same difference $c_y \lambda H(\bar r_y)$. Since $\lambda > 0$ and a nontrivial full-support distribution has positive entropy, $c_y = m_y$.

Locally, deterministic payoffs $o$ and $o^-$ have values $u(o)$ and zero by (8). Subtracting their two instances of (13), and identifying the resulting insertions with the two deterministic compounds in the statement, proves (12).

If $S_y$ is a singleton, append an independent full-support two-point dummy action to the first-stage action. The reached posterior becomes the product of $r_y$ and the dummy prior and hence has nontrivial support. The branch mass is still $m_y$, and Lemma 3 says that both compound values are unchanged. Apply the preceding argument in the product problem and remove the dummy. $\square$

**Lemma 6** (Deterministic branch assembly). *For every deterministic payoff profile $g : Y \to O$,*

$$W_q(C(P, g)) = \lambda I_q(A; Y) + \sum_y m_y u(g(y)). \tag{15}$$

*Proof.* First observe that the increment from changing one branch is independent of the payoffs on the other branches. Indeed, for any two backgrounds $g, h$, the two crossed public half-mixtures

$$\tfrac{1}{2}C(P, g[y \mapsto o]) \oplus \tfrac{1}{2}C(P, h[y \mapsto o^-])$$

and

$$\tfrac{1}{2}C(P, g[y \mapsto o^-]) \oplus \tfrac{1}{2}C(P, h[y \mapsto o])$$

have the same marked terminal law. Away from $y$ they contain the same two payoff-posterior atoms; at $y$ they contain $o$ and $o^-$ with the same mass $m_y/2$. Affinity of $W_q$ and cancellation of the two halves show that the two background-specific increments are equal. Taking $h$ to be the all-low profile and applying Lemma 5, then changing the finitely many branches one at a time, gives

$$W_q(C(P, g)) - W_q(C(P, o^-)) = \sum_y m_y u(g(y)). \tag{16}$$

The all-low compound pays $o^-$ surely and has exactly the information in the first-stage record. Lemma 4 therefore gives $W_q(C(P, o^-)) = \lambda I_q(A; Y)$. Substitute this into (16). □

### A.5 Proof of the theorem

*Proof of Theorem 1.* Assume first Axioms (A1)–(A8). Let $q$ have full support on a nontrivial action set, and let $K : A \to \Delta(O \times R)$ be arbitrary. Put $Y = O \times R$ and take as first stage

$$P_K(y \mid a) := K(y \mid a), \qquad y = (o, r).$$

After $y = (o, r)$, pay $o$ for sure and reveal nothing further. Call the resulting compound $K^{\text{seq}}$. From $K$, the record processor sends $(o, r)$ to the sequential record $((o, r), *)$. In the reverse direction it sends $((o, r), *)$ to $r$; inconsistent payoff-record pairs are unreached and may be completed arbitrarily. Both processors preserve the actual payoff coordinate. Hence Axiom (A4) gives $(q, K) \sim (q, K^{\text{seq}})$.

Apply Lemma 6 with $g(o, r) = o$. The two identities needed are

$$I_q(A; Y) = I_{qK}(A; O, R)$$

and

$$\sum_{o,r} p_{qK}(o, r) u(o) = \mathbb{E}_{q,K}[u(O)],$$

respectively. Together with sequentialisation, they give

$$W_q(K) = \mathbb{E}_{q,K}[u(O)] + \lambda I_{qK}(A; O, R). \tag{17}$$

If $A$ is a singleton, mutual information is zero and the marked terminal law is that of the induced payoff lottery. Equation (8) gives (17) directly. If $q$ is on the boundary, put $S = \operatorname{supp} q$ and write $(q^S, K^S)$ for the restriction. Then

$$(q, K) \sim (q^S, K^S), \qquad V(q, K) = V(q^S, K^S). \tag{18}$$

The first identity is Lemma 1(b); the second follows directly from the finite sums defining expected utility and mutual information. Thus no representative $W_q$ is assigned to a boundary fibre.

Restrict two arbitrary alternatives to their supports, apply (9) and (17) there, and use (18) to transport the comparison back. This gives

$$(q, K) \succeq (p, L) \quad \Longleftrightarrow \quad V(q, K) \geq V(p, L), \tag{19}$$

with the same $u$ and $\lambda$. Axiom (A3), first by duplication and then by deletion of irrelevant blocks, turns (19) into the within-channel representation and the finite-block moreover clause. The construction made $u$ nonconstant and $\lambda > 0$. This proves the forward implication.

Conversely, suppose that a nonconstant $u$ and a number $\lambda > 0$ represent the family by $V$. Axiom (A1) follows from the order on $\mathbb{R}$. Expected utility and mutual information are jointly continuous in $(K, q)$ on each finite simplex; for the latter, use the entropy formula and $0 \log 0 = 0$. Hence the set of triples $(K, q, p)$ for which $V(q, K) \geq V(p, K)$ is closed, which is Axiom (A2). A block-supported prior has a constant block label, which changes neither term of $V$; this proves Axiom (A3) and reduces every pair comparison below to the corresponding two values.

A payoff-preserving record processor leaves the payoff marginal unchanged and weakly lowers $I(A; O, R)$ by data processing. This proves Axiom (A4). Under an action report, equation (2) again leaves the payoff-record marginal unchanged, and the joint law forms the Markov chain

$$(O, R) - A - B.$$

Thus $I(B; O, R) \leq I(A; O, R)$, proving Axiom (A5). This argument uses the undivided joint-law equation and so also covers every completion on a null reported action.

For a compound channel, the two finite chain rules are

$$\mathbb{E}[u(O)] = \sum_{y:m(y)>0} m(y)\mathbb{E}_{q_y, K^y}[u(O)],$$
$$I(A; Y, O, R) = I(A; Y) + \sum_{y:m(y)>0} m(y) I_{q_y, K^y}(A; O, R).$$

The first-stage term is common to two continuation profiles. Their aggregate value difference is therefore

$$\sum_{y:m(y)>0} m(y)\big[V(q_y, K^y) - V(q_y, L^y)\big].$$

Branchwise weak improvement makes this sum nonnegative, and a strict improvement on a reached branch makes it positive. This is precisely Axiom (A6).

Because $u$ is nonconstant, two deterministic payoffs are strictly ranked; the trace term is zero on a singleton action set. This gives Axiom (A7). At a constant payoff, full revelation has value $u(o) + \lambda H(q)$ and an uninformative record has value $u(o)$. Full support on a nontrivial finite action set implies $H(q) > 0$; hence Axiom (A8). The same block calculation used above gives the moreover clause with the original witnesses $u$ and $\lambda$. □

## B The pure-trace characterization

Let $A$ be a finite set of *actions* and $X$ a finite set of *records*. A channel is a stochastic kernel $P : A \to \Delta(X)$. For each finite channel $P$, the primitive behavioural datum is a binary relation

$$\succeq_P \quad \text{on} \quad \Delta(A).$$

Condition (P1) below requires this relation to be a weak order. The interpretation is that $q \succeq_P q'$ means: in the fixed environment $P$, lottery $q$ displays at least as much traceability as $q'$. The environment $P$ is not an object of choice. A lottery is the procedure being evaluated; the channel is the fixed observation technology through which the realised act may be recorded by records.

**Comparison environments.** To compare lotteries carried by different channels without abandoning channel-indexed preferences, place the channels in labelled, disjoint blocks of one larger environment. For channels $P : A \to \Delta(X)$ and $Q : B \to \Delta(Y)$, write $P \sqcup Q$ for the block-diagonal channel with action alphabet $(A \times \{0\}) \cup (B \times \{1\})$ and record alphabet $(X \times \{0\}) \cup (Y \times \{1\})$. It sends $(a, 0)$ to $(x, 0)$ with probability $P(x \mid a)$ and $(b, 1)$ to $(y, 1)$ with probability $Q(y \mid b)$; all cross-block probabilities are zero. For $q \in \Delta(A)$ and $p \in \Delta(B)$, define lotteries on the tagged action alphabet by

$$\begin{aligned} q^0(a,0) &:= q(a), & q^0(b,1) &:= 0, \\ p^1(a,0) &:= 0, & p^1(b,1) &:= p(b), \end{aligned} \qquad (a \in A,\ b \in B).$$

The superscript is a block label, not a power. Accordingly, $q^0 \succeq_{P \sqcup Q} p^1$ is an ordinary preference comparison inside one fixed environment.[1] When the two blocks have the same action alphabet, $q^0$ and $q^1$ are the two tagged copies of the same lottery $q$. For a finite family $\{P_i : A_i \to \Delta(X_i)\}_{i \in K}$, write $\bigsqcup_{i \in K} P_i$ for the analogous finite block environment and $q_i^i$ for the lottery that assigns $q_i(a)$ to each tagged action $(a, i)$ and zero to every other block.

**Pure-record conditions.** The conditions are imposed directly on the family $\{\succeq_P\}_P$. Five conditions are imposed on the family. The first three are structural: each environment carries a continuous weak order with a fixed orientation (P1, P2), and labelled blocks provide the comparison framework (P3). The substantive restrictions are two: garbling cannot increase traceability (P4), and sequential observation is evaluated branch by branch (P5).

**(P1) Weak order and local non-triviality.** For every finite channel $P : A \to \Delta(X)$, $\succeq_P$ is complete and transitive on $\Delta(A)$. The family is locally non-trivial: for every finite action set $A$ with $|A| \geq 2$ and every full-support $q \in \Delta(A)$,

$$q^0 \succ_{\mathrm{Id}_A \sqcup U_A} q^1,$$

where $\mathrm{Id}_A : A \to \Delta(A)$ is full revelation and $U_A : A \to \Delta(\{*\})$ is the one-record uninformative channel.

This condition gives each observation environment an ordinary weak order over act lotteries and fixes the orientation of the comparison. When the realised act is genuinely uncertain, observing it directly is strictly more traceable than observing nothing about it.

[1] For a complete numerical example, let $A = \{a_1, a_2\}$, $B = \{b_1, b_2\}$, $X = \{x_0, x_1\}$, and $Y = \{y_0, y_1\}$. With rows indexed by actions and columns by records, take $P = \left(\begin{smallmatrix} 0.1 & 0.9 \\ 0.8 & 0.2 \end{smallmatrix}\right)$ and $Q = \left(\begin{smallmatrix} 0.4 & 0.6 \\ 0.6 & 0.4 \end{smallmatrix}\right)$. Order the tagged actions as $(a_1, 0), (a_2, 0), (b_1, 1), (b_2, 1)$ and the tagged records as $(x_0, 0), (x_1, 0), (y_0, 1), (y_1, 1)$. Then $P \sqcup Q = \mathrm{diag}(P, Q)$: the first two actions generate only the first two records, and the last two actions only the last two. If $q = (0.3, 0.7)$ and $p = (0.8, 0.2)$, their embeddings are $q^0 = (0.3, 0.7\,; 0, 0)$ and $p^1 = (0, 0\,; 0.8, 0.2)$, where the semicolon separates the two blocks. Thus $q^0 \succeq_{P \sqcup Q} p^1$ says precisely that lottery $q$, observed through $P$, is at least as traceable as lottery $p$, observed through $Q$; it does not treat either channel as an object of choice. Under the representation proved below, the two sides yield approximately 0.330 and 0.019 bits of mutual information, respectively, and hence the comparison is strict in this example.

**(P2) Continuity.** For every pair of finite alphabets $(A, X)$, the set

$$\{(P, q, q') \in \Delta(X)^A \times \Delta(A) \times \Delta(A) : q \succeq_P q'\}$$

is closed in the usual Euclidean product topology. Thus within-environment rankings vary continuously with both the lotteries and the fixed channel.

Continuity says that traceability comparisons are stable under small changes in the procedure and the environment. Small modelling perturbations should not create abrupt reversals. The condition is stated entirely in the primitives: it quantifies over channels and lotteries, not over derived Bayesian objects. Continuity of block comparisons in posterior-law convergence at a fixed full-support prior is not assumed; it is derived below (Lemma 14).

**(P3) Block-comparison coherence.** *Duplication:* for every channel $P : A \to \Delta(X)$ and every $q, q' \in \Delta(A)$,

$$q \succeq_P q' \iff q^0 \succeq_{P \sqcup P} (q')^1.$$

*Irrelevant blocks:* for every finite block environment $\bigsqcup_{k \in K} P_k$, every ordered pair of distinct blocks $i, j \in K$, and every $q_i \in \Delta(A_i)$ and $q_j \in \Delta(A_j)$,

$$q_i^i \succeq_{\bigsqcup_{k \in K} P_k} q_j^j \iff q_i^0 \succeq_{P_i \sqcup P_j} q_j^1.$$

Applied at the ordered pair $(1, 0)$ of a two-block environment, the second clause exchanges block labels; this is how reversed block comparisons are brought to canonical form throughout (Lemma 10(a)).

Labelled blocks are a way to place procedures carried by different channels in a common comparison problem. The labels should not themselves create traceability, alter the ranking inside a duplicated environment, or let unused blocks affect the comparison between active blocks.

**(P4) Split data-processing aversion.** Condition (P4) is the conjunction of the following two clauses. This is the formulation used by the formal theorem: the record and action coordinates are processed separately, and no convention is imposed on null reported actions.

(DP-rec) *Record post-processing.* For every channel $P : A \to \Delta(X)$, every stochastic $T : X \to \Delta(X')$, and every $q \in \Delta(A)$, define

$$(PT)(x' \mid a) := \sum_{x \in X} P(x \mid a) T(x' \mid x).$$

Then

$$q^0 \succeq_{P \sqcup PT} q^1.$$

(DP-act) *Bayesian action coarsening.* For every $P : A \to \Delta(X)$, every stochastic $S : A \to \Delta(A')$, every $q \in \Delta(A)$, write

$$(qS)(a') := \sum_{a \in A} q(a) S(a' \mid a).$$

For every channel $\widehat{P} : A' \to \Delta(X)$ satisfying

$$(qS)(a') \widehat{P}(x \mid a') = \sum_{a \in A} q(a) P(x \mid a) S(a' \mid a) \quad \text{for every } (a', x) \in A' \times X, \quad \text{(DP-act-law)}$$

we have

$$q^0 \succeq_{P \sqcup \widehat{P}} (qS)^1.$$

When $qS$ has full support, (DP-act-law) determines $\widehat{P}$ uniquely; otherwise the clause applies to every channel consistent with it. In particular, bijective relabellings are neutral by applying (DP-act) in both directions.

The act whose trace is evaluated may be reported through a possibly noisy description, and the visible record may be garbled after it is generated. Neither operation can create observable trace: the original procedure is weakly more traceable than any procedure obtained by either operation.

**Remark 7.** The two primitive clauses imply the convenient simultaneous form used in a few calculations. Let $S : A \to \Delta(A')$ and $T : X \to \Delta(X')$, and let $P' : A' \to \Delta(X')$ satisfy

$$(qS)(a')\, P'(x' \mid a') = \sum_{a\in A} \sum_{x\in X} q(a)P(x \mid a)S(a' \mid a)T(x' \mid x) \quad \text{for every } (a', x') \in A' \times X'. \tag{DP}$$

First apply (DP-rec) to form $PT$, and then apply (DP-act) to $(q, PT)$ through $S$. Equation (DP) says exactly that the supplied $P'$ is an allowed Bayesian completion of this second step, including on null reported actions. Chain the two comparisons in a common three-block environment using Conditions (P1) and (P3). This yields $q^0 \succeq_{P\sqcup P'} (qS)^1$. Conversely, the simultaneous form specialises to each named clause. Thus the two presentations are equivalent under (P1) and (P3), but the proposition below takes the two named clauses as its literal premise. If $(qS)(a') = 0$, both (DP-act-law) and (DP) leave the corresponding channel row unrestricted; null reports require no convention.

**Reached records and posteriors.** For a lottery $q \in \Delta(A)$ and a channel $P : A \to \Delta(X)$, write

$$m_{q,P}(x) := \sum_{a\in A} q(a)P(x \mid a), \qquad r_x^{q,P}(a) := \frac{q(a)P(x \mid a)}{m_{q,P}(x)} \quad \text{if } m_{q,P}(x) > 0$$

for, respectively, the probability of record $x$ and the posterior belief about the realised action. Call $x$ reached when $m_{q,P}(x) > 0$; all branch comparisons below are restricted to reached records.

**(P5) Branchwise continuation monotonicity.** Let $P_1 : A \to \Delta(X_1)$ be a first-stage channel and, after each first-stage record $x$, let $Q^x, R^x : A \to \Delta(X_2^x)$ be two alternative continuation channels. Write $P_1 \triangleright \{Q^x\}$, and analogously $P_1 \triangleright \{R^x\}$, for the compound channel with record alphabet $\bigcup_{x\in X_1}(\{x\} \times X_2^x)$ that reports both stages:

$$(P_1 \triangleright \{Q^x\})(x, z \mid a) := P_1(x \mid a)Q^x(z \mid a).$$

Fix $q \in \Delta(A)$ and let $m(x) := m_{q,P_1}(x)$ and $r_x := r_x^{q,P_1}$ for all branches with $m(x) > 0$. If

$$r_x^0 \succeq_{Q^x\sqcup R^x} r_x^1 \qquad \text{for every } x \text{ with } m(x) > 0,$$

then

$$q^0 \succeq_{(P_1\triangleright\{Q^x\})\sqcup(P_1\triangleright\{R^x\})} q^1.$$

If at least one reached branch comparison is strict, then the aggregate comparison is strict.

Sequential observation is evaluated branch by branch. After a first-stage signal, if one continuation leaves at least as much trace as another at every posterior that can be reached, then replacing the continuation branch by branch should not reduce overall traceability. If the improvement is strict on a reached branch, the aggregate comparison is strict.

**Remark 8.** Condition (P5) plays in this system the role the sure-thing principle plays in Savage's: it is a dominance condition across the branches of a resolving observation, with zero-probability branches unconstrained exactly as Savage's null events are. It is also where all cardinal content resides. The other conditions are ordinal, environment-by-environment conditions; it is (P5), through its derived consequences—public-coin independence (Lemma 13), background inertness (Lemma 20), and the chain rule (Lemma 25)—that aligns the initially separate scales across priors and forces branch-additive aggregation. The strictness clause is not decorative: it is what allows the converse directions of Lemmas 13 and 20 to be extracted from a dominance condition. For example, a mutual-information index capped at a fixed level satisfies (P1)–(P4) but fails exactly (P5). Compound observation environments are accordingly the natural locus of experimental testing: they are where branch-additive and non-additive criteria come apart.

### B.1 Auxiliary result

Throughout, logarithms are base two, so information is measured in bits; changing the base merely rescales all information values. For a finite distribution $x$, write

$$H(x) := -\sum_{z} x(z)\log x(z), \qquad 0\log 0 := 0,$$

and, for a prior–channel pair $(q, P)$, define

$$I_{q,P}(A;X) := H(m_{q,P}) - \sum_{a\in A} q(a)H(P(\cdot \mid a)).$$

When the alphabets are clear, write $I_{q,P}$; write $I$ for the map $(q,P) \mapsto I_{q,P}$.

**Proposition 9** (Pure-trace characterisation)**.** *For a family $\{\succeq_P\}_P$ as described above, the following are equivalent.*

(i) *The family satisfies Conditions* (P1)–(P5)*.*

(ii) *For every finite channel $P : A \to \Delta(X)$ and every $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \quad\Longleftrightarrow\quad I_{q,P}(A;X) \geq I_{q',P}(A;X).$$

*Moreover, under these equivalent conditions, block-supported cross-channel comparisons are represented on the same scale: for every finite block environment $\bigsqcup_{k\in K} P_k$, every two distinct blocks $i, j \in K$, and every $q_i \in \Delta(A_i)$, $q_j \in \Delta(A_j)$,*

$$q_i^i \succeq_{\bigsqcup_{k\in K} P_k} q_j^j \quad\Longleftrightarrow\quad I_{q_i,P_i}(A_i;X_i) \geq I_{q_j,P_j}(A_j;X_j).$$

### B.2 Proof of the pure-trace characterization

#### B.2.1 Necessity

For every finite channel $P : A \to \Delta(X)$, define

$$q \succeq_P^I q' \quad\Longleftrightarrow\quad I_{q,P}(A;X) \geq I_{q',P}(A;X).$$

We verify that the family $\{\succeq_P^I\}_P$ satisfies Conditions (P1)–(P5): this is the implication (ii)⇒(i) of Proposition 9, and it establishes the consistency of the condition system.

The *posterior law* induced by $(q, P)$ is the finite distribution

$$\mu_{q,P} := \sum_{x:m_{q,P}(x)>0} m_{q,P}(x)\,\delta_{r_x^{q,P}}$$

over posterior beliefs. Bayes plausibility gives $\int r\,d\mu_{q,P}(r) = q$. We repeatedly use the elementary identities

$$I_{q,P} = H(m_{q,P}) - \sum_{a\in A} q(a)H(P(\cdot \mid a)), \tag{N1}$$

and

$$I_{q,P} = H(q) - \int_{\Delta(A)} H(r)\, d\mu_{q,P}(r). \tag{N2}$$

Both formulae are read on the support of the relevant distributions; zero-probability rows and records do not affect the value.

*Condition* (P1). For each fixed $P$, the relation $\succeq^I_P$ is complete and transitive because it is represented by the real-valued function $q \mapsto I_{q,P}$. If $|A| \geq 2$ and $q$ has full support, then full revelation gives

$$I_{q,\mathrm{Id}_A} = H(q) > 0,$$

whereas the one-record uninformative channel $U_A$ gives $I_{q,U_A} = 0$. Hence

$$q^0 \succ^I_{\mathrm{Id}_A \sqcup U_A} q^1.$$

*Condition* (P2). For fixed finite alphabets, (N1) shows that $(P, q) \mapsto I_{q,P}$ is continuous, since Shannon entropy is continuous on every finite simplex. Therefore the set

$$\{(P, q, q') : q \succeq^I_P q'\} = \{(P, q, q') : I_{q,P} \geq I_{q',P}\}$$

is closed. Posterior-law continuity of block comparisons is not an condition; for the mutual-information family it also holds directly, by (N2) and continuity of $H$ on the compact simplex $\Delta(A)$, consistently with its derivation from the conditions in Lemma 14 below.

*Condition* (P3). For every channel $P$ and lotteries $q, q' \in \Delta(A)$,

$$I_{q^0,P\sqcup P} = I_{q,P}, \qquad I_{(q')^1,P\sqcup P} = I_{q',P},$$

because a block-supported lottery puts probability one on its own block and the block label is constant under that lottery. Hence

$$q \succeq^I_P q' \iff q^0 \succeq^I_{P\sqcup P} (q')^1.$$

The same observation proves invariance to adding or deleting irrelevant blocks. If $q^i_i$ and $q^j_j$ are supported on blocks $i$ and $j$, their mutual-information values in $\bigsqcup_{k\in K} P_k$ are exactly

$$I_{q_i,P_i} \quad \text{and} \quad I_{q_j,P_j},$$

independently of all other blocks.

*Condition* (P4). It is useful to verify the stronger simultaneous inequality from Remark 7, which immediately supplies both named clauses. Let $S : A \to \Delta(A')$ and $T : X \to \Delta(X')$ be stochastic, and let $P'$ satisfy (DP). Generate $A \sim q$; then, conditionally independently given $A$, generate $A'$ from $S(\cdot \mid A)$ and $X$ from $P(\cdot \mid A)$; finally generate $X'$ from $T(\cdot \mid X)$. The joint law of $(A, A', X, X')$ is

$$q(a)S(a' \mid a)P(x \mid a)T(x' \mid x),$$

so $A'$ is conditionally independent of $(X, X')$ given $A$, and $X'$ is conditionally independent of $(A, A')$ given $X$. Hence

$$A' \longleftarrow A \longrightarrow X \longrightarrow X'$$

is a Markov chain in the sense that $A' \to A \to X \to X'$ satisfies the required conditional independences. Data processing applied to the fork $A' \leftarrow A \to X$ gives $I(A; X) \geq I(A'; X)$, and data processing applied to $A' \to X \to X'$ gives $I(A'; X) \geq I(A'; X')$. Therefore

$$I_{q,P}(A; X) \geq I(A'; X').$$

By construction the joint law of $(A', X')$ is exactly $(qS)(a')P'(x' \mid a')$, the left-hand side of (DP), so $I(A'; X') = I_{qS,P'}(A'; X')$. Rows $a'$ with $(qS)(a') = 0$ carry no mass under $qS$, so this value is the same for every $P'$ satisfying (DP), however the null rows are completed. Using the block computation of (P3) above to evaluate the two blocks separately,

$$q^0 \succeq^I_{P \sqcup P'} (qS)^1.$$

The first named clause is the case $S = \mathrm{Id}_A$ and gives $I_{q,P}(A;X) \geq I_{q,PT}(A;X')$. For later use, in the action-only case $T = \mathrm{Id}_X$ write $S^qP$ for the conditional channel from the reported action to the record on every reached row:

$$S^qP(x \mid a') := \frac{\sum_{a \in A} q(a)S(a' \mid a)P(x \mid a)}{(qS)(a')} \qquad \text{when } (qS)(a') > 0.$$

A *completion* of $S^qP$ is any channel $\widehat{P} : A' \to \Delta(X)$ that agrees with this formula on reached rows; its values on null rows are unrestricted. The second named clause gives $I_{q,P}(A;X) \geq I_{qS,\widehat{P}}(A';X)$ for every completion $\widehat{P}$ of $S^qP$.

*Condition* (P5). Fix a first-stage channel $P_1 : A \to \Delta(X_1)$. Let

$$m(x) = m_{q,P_1}(x), \qquad r_x = r_x^{q,P_1}.$$

For a continuation profile $\mathcal{Q} = \{Q^x\}_x$, the chain rule for mutual information gives

$$I_{q,P_1 \rhd \mathcal{Q}}(A; X_1, X_2) = I_{q,P_1}(A;X_1) + \sum_{x:m(x)>0} m(x)\, I_{r_x,Q^x}(A; X_2^x). \tag{N3}$$

Therefore, if

$$r_x^0 \succeq^I_{Q^x \sqcup R^x} r_x^1 \quad \text{for every positive-probability branch } x,$$

then $I_{r_x,Q^x} \geq I_{r_x,R^x}$ for every such $x$. Using (N3),

$$I_{q,P_1 \rhd \{Q^x\}} \geq I_{q,P_1 \rhd \{R^x\}},$$

so

$$q^0 \succeq^I_{(P_1 \rhd \{Q^x\}) \sqcup (P_1 \rhd \{R^x\})} q^1.$$

If one branch comparison is strict and that branch has positive probability, then the corresponding summand in (N3) is strictly larger, so the aggregate comparison is strict.

No product condition is imposed, so the verification ends here. For later use, write $q_1 \otimes q_2$ for the product lottery and define the product of channels by

$$(P_1 \otimes P_2)(x_1, x_2 \mid a_1, a_2) := P_1(x_1 \mid a_1)P_2(x_2 \mid a_2).$$

The mutual-information family satisfies the background-inertness property derived in Lemma 20 below, since

$$I_{q_1 \otimes q_2, P_1 \otimes R_2}(A_1, A_2; X_1, X_2) = I_{q_1,P_1}(A_1;X_1) + I_{q_2,R_2}(A_2;X_2), \tag{N4}$$

so that the comparison of $P_1 \otimes R_2$ and $Q_1 \otimes R_2$ at $q_1 \otimes q_2$ reduces to the comparison of $P_1$ and $Q_1$ at $q_1$, whatever the common background $R_2$. Consistently with Lemma 20, this is a consequence of the chain rule (N3) and not an independent requirement.

### B.2.2 Sufficiency

Assume Conditions (P1)–(P5). We prove the implication (i)⇒(ii) in Proposition 9. The proof is written first for full-support lotteries. Boundary cases are obtained by exact restriction to the positive support and transport back to the original action set. The key point is that posterior-law sufficiency is not assumed; it is derived from block comparisons and record post-processing, and so is its continuity (Lemma 14).

**How to read the proof.** The argument has five interfaces. Each stage ends with exactly the object needed by the next one:

Stage 1: replace channels by Bayes-plausible posterior laws and derive the needed quotient continuity from primitive closedness;

Stage 2: cardinalise each fixed-prior quotient and make support, neutrality, and relabelling behaviour explicit;

Stage 3: transport those affine values through independent products and into one cross-prior block scale, allowing provisionally a bilinear interaction;

Stage 4: identify reached-branch coefficients, prove their cocycle, and align the scales on nested support faces;

Stage 5: compare product and sequential decompositions, eliminate the interaction, and invoke Faddeev's recursion to obtain Shannon entropy and hence mutual information.

The long posterior-grid, product-coefficient, and branch tangent-space calculations are technical subarguments. A reader interested in the logical spine may use their lemma statements as interfaces. The kernel proof chooses its canonical representative before the branch and product calculations and implements the branch interfaces using finite atomic signed posterior laws. The paper keeps product calibration before branch calibration because this makes the scalar algebra easier to read. The early full-revelation normalisation and exact relabelling lemma below supply the coherence needed by that presentation; the kernel proof reaches the same later interfaces in the opposite order.

**Stage 1: posterior-law quotient and derived continuity.** The first five lemmas establish a coherent order on posterior laws. In particular, posterior-law sufficiency and continuity are conclusions, not extra axioms; the barycentric spread–merge construction in Lemma 14 is the technical core of this stage.

**Lemma 10** (Coherent block comparisons)**.** *Assume Conditions* (P1) *and* (P3).

(a) *Let* $(q_1, P_1)$, $(q_2, P_2)$, $(q_3, P_3)$ *be pairs of a lottery* $q_k \in \Delta(A_k)$ *and a channel* $P_k : A_k \to \Delta(X_k)$, *with action sets, record sets, and lotteries not necessarily equal and the lotteries not necessarily of full support. Then the two-block comparisons between such pairs are complete,*

$$q_1^0 \succeq_{P_1 \sqcup P_2} q_2^1 \quad \textit{or} \quad q_2^0 \succeq_{P_2 \sqcup P_1} q_1^1,$$

*satisfy the reverse-block equivalence*

$$q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0 \quad \Longleftrightarrow \quad q_2^0 \succeq_{P_2 \sqcup P_1} q_1^1,$$

*and are transitive:*

$$q_1^0 \succeq_{P_1 \sqcup P_2} q_2^1 \quad \textit{and} \quad q_2^0 \succeq_{P_2 \sqcup P_3} q_3^1 \quad \Longrightarrow \quad q_1^0 \succeq_{P_1 \sqcup P_3} q_3^1.$$

*In particular, for a fixed full-support* $q \in \Delta(A)$ *the comparisons* $q^0 \succeq_{P \sqcup Q} q^1$ *define a weak order over channels on* $A$ *at prior* $q$*, and pairwise comparisons between block-supported lotteries are independent of the block environment in which the two blocks are embedded.*

(b) *Under* (P2), *the comparisons in* (a) *are also closed in the channels at fixed alphabets and fixed lotteries: if* $P_n \to P$ *in* $\Delta(X_1)^{A_1}$, $Q_n \to Q$ *in* $\Delta(X_2)^{A_2}$, $q_1 \in \Delta(A_1)$ *and* $q_2 \in \Delta(A_2)$ *are fixed, and*

$$q_1^0 \succeq_{P_n \sqcup Q_n} q_2^1 \quad \textit{for all } n,$$

*then*

$$q_1^0 \succeq_{P \sqcup Q} q_2^1.$$

*Fixedness of the alphabets is inherited from Condition* (P2), *which is stated at a fixed pair of alphabets.*

*Proof.* (a) Completeness follows from Condition (P1) applied to the two-block environment $P_1 \sqcup P_2$: either $q_1^0 \succeq_{P_1 \sqcup P_2} q_2^1$ or $q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0$. In the second case, applying Condition (P3) to the two-block environment with blocks $P_1$ and $P_2$, using the ordered pair $(i, j) = (1, 0)$, gives the canonical reverse comparison

$$q_2^1 \succeq_{P_1 \sqcup P_2} q_1^0 \iff q_2^0 \succeq_{P_2 \sqcup P_1} q_1^1,$$

which is the reverse-block equivalence.

For transitivity, form the three-block environment $\bigsqcup_{k=1}^3 P_k$. By the irrelevant-blocks clause of Condition (P3) with block pair $(1, 2)$, the first hypothesis is equivalent to

$$q_1^1 \succeq_{\bigsqcup_{k=1}^3 P_k} q_2^2,$$

and by the same clause with block pair $(2, 3)$, the second hypothesis is equivalent to

$$q_2^2 \succeq_{\bigsqcup_{k=1}^3 P_k} q_3^3.$$

Transitivity of the weak order given by Condition (P1) in this common environment gives

$$q_1^1 \succeq_{\bigsqcup_{k=1}^3 P_k} q_3^3,$$

and applying Condition (P3) again, now deleting the irrelevant middle block, yields $q_1^0 \succeq_{P_1 \sqcup P_3} q_3^1$. Nothing in these arguments requires the pairs to share alphabets or lotteries, or the lotteries to have full support. The specialisation to a fixed full-support $q$ and channels on a common $A$ is immediate.

(b) The block channel $P_n \sqcup Q_n$ has fixed tagged copies of the alphabets $A_1, A_2, X_1, X_2$ and converges to $P \sqcup Q$ in the finite-dimensional product simplex. Since the block-supported lotteries $q_1^0, q_2^1$ are fixed, Condition (P2) applied at this pair of alphabets gives

$$q_1^0 \succeq_{P \sqcup Q} q_2^1. \qquad \square$$

**Lemma 11** (Prior-specific Blackwell equivalence)**.** *Let $q \in \Delta(A)$ have full support. Let $P : A \to \Delta(X)$ and $Q : A \to \Delta(Y)$ be finite channels. If*

$$\mu_{q,P} = \mu_{q,Q},$$

*then there are stochastic kernels $T : X \to \Delta(Y)$ and $T' : Y \to \Delta(X)$ such that*

$$Q = PT, \qquad P = QT'.$$

*Proof.* It is enough to construct $T$; the construction of $T'$ is symmetric. Let

$$m_P(x) := m_{q,P}(x), \qquad m_Q(y) := m_{q,Q}(y).$$

If $m_P(x) = 0$, then $P(x \mid a) = 0$ for every $a \in A$, because $q$ has full support. Such records do not affect $PT$, so define the corresponding row of $T$ arbitrarily. The same observation applies to null records of $Q$.

Let $\mathcal{R}$ be the finite set of posterior vectors that occur with positive probability under either channel. For $r \in \mathcal{R}$, define

$$X_r := \{x \in X : m_P(x) > 0,\ r_x^{q,P} = r\}, \qquad Y_r := \{y \in Y : m_Q(y) > 0,\ r_y^{q,Q} = r\}.$$

The sets $\{X_r\}_{r \in \mathcal{R}}$ partition the positive-probability records of $P$, and the sets $\{Y_r\}_{r \in \mathcal{R}}$ partition the positive-probability records of $Q$, by posterior vector. Equality of posterior laws implies that the total probability attached to posterior $r$ is the same for the two channels:

$$\lambda_r := \sum_{x \in X_r} m_P(x) = \sum_{y \in Y_r} m_Q(y) > 0.$$

For $x \in X_r$ and $y \in Y_r$, set

$$T(y \mid x) := \frac{m_Q(y)}{\lambda_r},$$

and set $T(y \mid x) = 0$ when $x \in X_r$ and $y \notin Y_r$. This defines a stochastic row for every positive-probability record $x$, since the probabilities over $Y_r$ sum to one. Rows corresponding to null records of $P$ were already chosen arbitrarily.

Fix $a \in A$ and $y \in Y_r$. Since all records in $X_r$ have posterior $r$, Bayes' rule gives

$$q(a)P(x \mid a) = m_P(x)r(a) \qquad (x \in X_r).$$

Therefore

$$\sum_{x \in X_r} P(x \mid a) = \frac{\lambda_r r(a)}{q(a)}.$$

Using the definition of $T$,

$$(PT)(y \mid a) = \sum_{x \in X_r} P(x \mid a)\frac{m_Q(y)}{\lambda_r} = \frac{m_Q(y)r(a)}{q(a)}.$$

But $y \in Y_r$ also has posterior $r$, so Bayes' rule for $Q$ gives

$$Q(y \mid a) = \frac{m_Q(y)r(a)}{q(a)}.$$

Thus $(PT)(y \mid a) = Q(y \mid a)$ for every positive-probability $y$. If $m_Q(y) = 0$, then full support of $q$ implies $Q(y \mid a) = 0$ for every $a$. No positive-probability record of $P$ sends mass to such a $y$ under the construction; rows of $T$ indexed by null records of $P$ may be chosen arbitrarily, but those records have $P(x \mid a) = 0$ for every $a$, so they contribute nothing to $PT$. Hence $(PT)(y \mid a) = 0 = Q(y \mid a)$ for every $a$, and therefore $Q = PT$. □

**Lemma 12** (Posterior-law sufficiency)**.** *Fix a full-support $q \in \Delta(A)$. If $P : A \to \Delta(X)$ and $Q : A \to \Delta(Y)$ generate the same posterior law at $q$,*

$$\mu_{q,P} = \mu_{q,Q},$$

*then*

$$q^0 \sim_{P \sqcup Q} q^1.$$

*Consequently, by Lemma 10, at a fixed full-support prior the block comparison of channels depends only on the induced posterior law.*

*Proof.* By Lemma 11, there are stochastic kernels $T$ and $T'$ such that $Q = PT$ and $P = QT'$. Clause (DP-rec) of Condition (P4) gives

$$q^0 \succeq_{P \sqcup PT} q^1, \qquad q^0 \succeq_{Q \sqcup QT'} q^1.$$

Using $PT = Q$, the first comparison is

$$q^0 \succeq_{P \sqcup Q} q^1.$$

Using $QT' = P$, the second comparison is

$$q^0 \succeq_{Q \sqcup P} q^1.$$

By the reverse-block equivalence in Lemma 10 (equivalently, Condition (P3) applied to $P \sqcup Q$ with ordered pair $(i, j) = (1, 0)$), this is equivalent to

$$q^1 \succeq_{P \sqcup Q} q^0.$$

Thus $q^0 \sim_{P \sqcup Q} q^1$. □

For channels $P : A \to \Delta(X_P)$ and $R : A \to \Delta(X_R)$ and $\lambda \in [0, 1]$, write $\lambda P \oplus (1-\lambda)R$ for the channel that publicly selects $P$ with probability $\lambda$ and $R$ otherwise, and reports the selected channel together with its record. Thus

$$(\lambda P \oplus (1-\lambda)R)((z,i) \mid a) = \begin{cases} \lambda P(z \mid a), & i = 0, \\ (1-\lambda)R(z \mid a), & i = 1. \end{cases}$$

Because the coin is observed,

$$\mu_{q,\lambda P \oplus (1-\lambda)R} = \lambda \mu_{q,P} + (1-\lambda)\mu_{q,R}.$$

**Lemma 13** (Derived public-coin independence)**.** *Assume Conditions* (P1) *and* (P3)*, clause* (DP-rec) *of Condition* (P4)*, and Condition* (P5)*. Fix a finite action set $A$, a full-support $q \in \Delta(A)$, and channels $P, Q, R$ on $A$. For every $\lambda \in (0, 1)$,*

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \Longleftrightarrow \quad q^0 \succeq_{(\lambda P \oplus (1-\lambda)R) \sqcup (\lambda Q \oplus (1-\lambda)R)} q^1.$$

*Proof.* Throughout, for channels $V, W$ on $A$ write $V \succeq_q W$ for the block comparison $q^0 \succeq_{V \sqcup W} q^1$, and $V \succ_q W$ for the corresponding strict comparison. By Lemma 10 the comparison $\succeq_q$ is complete and transitive, and by Lemma 12 it depends only on the pair of posterior laws $(\mu_{q,V}, \mu_{q,W})$. In particular, channels with equal posterior laws at $q$ may be substituted on either side of any comparison; we refer to this as *substitution.* Substitution also preserves strict comparisons: if $\mu_{q,V} = \mu_{q,V'}$ and $\mu_{q,W} = \mu_{q,W'}$, then $V \sim_q V'$ and $W \sim_q W'$ by Lemma 12, and transitivity of the complete relation $\succeq_q$ gives $V \succ_q W$ if and only if $V' \succ_q W'$.

*Step 1: the public coin is an uninformative first stage.* Define the constant channel $C : A \to \Delta(\{0, 1\})$ by

$$C(0 \mid a) := \lambda, \qquad C(1 \mid a) := 1 - \lambda \qquad (a \in A).$$

Since $\lambda \in (0, 1)$, both branches have positive mass: $m_{q,C}(0) = \lambda$ and $m_{q,C}(1) = 1 - \lambda$. Because the rows of $C$ are constant, Bayes' rule gives, for each $x \in \{0, 1\}$,

$$r_x^{q,C}(a) = \frac{q(a)C(x \mid a)}{m_{q,C}(x)} = q(a),$$

so both branch posteriors equal the prior: $r_0 = r_1 = q$.

*Step 2: padding the continuations to a common alphabet.* Condition (P5) requires the two continuation channels on a given branch to share a record alphabet, whereas $P$ and $Q$ may not. Let $X_{PQ} := X_P \sqcup X_Q$ and define $\widetilde{P}, \widetilde{Q} : A \to \Delta(X_{PQ})$ by

$$\widetilde{P}(z \mid a) := \begin{cases} P(z \mid a) & z \in X_P, \\ 0 & z \in X_Q, \end{cases} \qquad \widetilde{Q}(z \mid a) := \begin{cases} 0 & z \in X_P, \\ Q(z \mid a) & z \in X_Q. \end{cases}$$

Padding adds only zero-probability records, so the positive-mass records and their posteriors are unchanged: $\mu_{q,\widetilde{P}} = \mu_{q,P}$ and $\mu_{q,\widetilde{Q}} = \mu_{q,Q}$. By substitution,

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \Longleftrightarrow \quad \widetilde{P} \succeq_q \widetilde{Q},$$

and the same equivalence holds for the strict comparisons.

*Step 3: the compounds implement the public-coin mixtures.* Consider the two-stage channels with first stage $C$, continuation pair $\widetilde{P}$ versus $\widetilde{Q}$ on branch 0, and the common continuation $R$ on branch 1:

$$C \triangleright \{\widetilde{P}, R\} \qquad \text{and} \qquad C \triangleright \{\widetilde{Q}, R\},$$

where the profile notation lists the branch-0 and branch-1 continuations in order. By the definition of two-stage composition,

$$(C \triangleright \{\widetilde{P}, R\})(0, z \mid a) = \lambda\, \widetilde{P}(z \mid a), \qquad (C \triangleright \{\widetilde{P}, R\})(1, z \mid a) = (1-\lambda)\, R(z \mid a).$$

A record $(0, z)$ has marginal mass $\lambda\, m_{q,\widetilde{P}}(z)$, and when this is positive the constant factor $\lambda$ cancels in Bayes' rule, so its posterior is $r_z^{q,\widetilde{P}}$; symmetrically for records $(1, z)$. Hence

$$\mu_{q,C\triangleright\{\widetilde{P},R\}} = \lambda\mu_{q,\widetilde{P}} + (1-\lambda)\mu_{q,R} = \lambda\mu_{q,P} + (1-\lambda)\mu_{q,R} = \mu_{q,\lambda P\oplus(1-\lambda)R},$$

using the public-coin mixing identity, and likewise $\mu_{q,C\triangleright\{\widetilde{Q},R\}} = \mu_{q,\lambda Q\oplus(1-\lambda)R}$. By substitution, the right-hand comparison of the lemma is therefore equivalent to

$$C \triangleright \{\widetilde{P}, R\} \succeq_q C \triangleright \{\widetilde{Q}, R\}. \tag{$*$}$$

*Step 4: forward implication.* Assume $q^0 \succeq_{P\sqcup Q} q^1$. We verify the branch hypotheses of Condition (P5) for the first stage $C$ and the continuation profiles $\{\widetilde{P}, R\}$ versus $\{\widetilde{Q}, R\}$. Both branches have positive mass and both branch posteriors equal $q$ (Step 1). On branch 0 the hypothesis is $q^0 \succeq_{\widetilde{P}\sqcup\widetilde{Q}} q^1$, which holds by Step 2. On branch 1 the two profiles share the continuation $R$, and the hypothesis $q^0 \succeq_{R\sqcup R} q^1$ holds because Condition (P1) gives $q \succeq_R q$ and the duplication clause of Condition (P3) converts it into the two-block form. Condition (P5) then yields $(*)$, hence the mixture comparison by Step 3.

*Step 5: converse, via the strictness clause.* Assume the mixture comparison holds, so that $(*)$ holds by Step 3, and suppose towards a contradiction that $q^0 \succeq_{P\sqcup Q} q^1$ fails. By completeness and the reverse-block equivalence of Lemma 10, $Q \succ_q P$, and Step 2 gives $\widetilde{Q} \succ_q \widetilde{P}$. Now apply Condition (P5) with the roles of the two profiles exchanged: the branch-0 comparison $\widetilde{Q} \succ_q \widetilde{P}$ is strict on a branch of mass $\lambda > 0$, and the branch-1 comparison $R \succeq_q R$ holds as in Step 4. The strictness clause of Condition (P5) yields

$$C \triangleright \{\widetilde{Q}, R\} \succ_q C \triangleright \{\widetilde{P}, R\},$$

which contradicts $(*)$. Hence $q^0 \succeq_{P\sqcup Q} q^1$. □

The lemma shows that public-coin independence is not an independent behavioural commitment: a public coin is exactly a first-stage observation that reveals nothing about the act, so invariance under common public mixtures is branchwise continuation monotonicity applied through trivial branch posteriors. The statement is given at full-support priors, which is the only case used below (Lemma 15).

For a finite action set $A$, let $\mathsf{X}_A$ be the set of finite-support probability measures on $\Delta(A)$. Write

$$b(\mu) := \int_{\Delta(A)} r\, d\mu(r), \qquad \mathcal{M}_q := \{\mu \in \mathsf{X}_A : b(\mu) = q\}.$$

Thus $\mathcal{M}_q$ is the set of finite posterior laws with prior $q$ as their mean, and $\mu_{q,P} \in \mathcal{M}_q$ for every channel $P$.

**Lemma 14** (Posterior-law continuity)**.** *Fix a full-support $q \in \Delta(A)$. Let $P_n, Q_n, P, Q$ be finite channels on $A$ with*

$$\mu_{q,P_n} \Rightarrow \mu_{q,P}, \qquad \mu_{q,Q_n} \Rightarrow \mu_{q,Q}.$$

*If*

$$q^0 \succeq_{P_n\sqcup Q_n} q^1 \qquad \textit{for all } n,$$

*then*

$$q^0 \succeq_{P\sqcup Q} q^1.$$

*Proof.* Throughout the proof, for channels $V, W$ on $A$ write $V \succeq_q W$ for the block comparison $q^0 \succeq_{V\sqcup W} q^1$. By Lemma 10 this comparison is complete and transitive, and by Lemma 12 it depends only on the pair of posterior laws $(\mu_{q,V}, \mu_{q,W})$; in particular, channels with equal posterior laws at $q$ may be substituted on either side of any comparison. We refer to this as *substitution.*

For a finite-support Bayes-plausible law

$$\mu = \sum_{k\in K} \lambda_k \delta_{r_k} \in \mathcal{M}_q,$$

with atoms listed so that $\lambda_k > 0$, define the canonical implementing channel $C_\mu : A \to \Delta(K)$ by

$$C_\mu(k \mid a) := \frac{\lambda_k r_k(a)}{q(a)}.$$

Full support of $q$ and $\sum_k \lambda_k r_k = q$ make $C_\mu$ a channel with $\mu_{q,C_\mu} = \mu$.

*Step 1: merging posteriors moves weakly down.* Let $\mu = \sum_{k\in K} \lambda_k \delta_{r_k} \in \mathcal{M}_q$ and let $\{K_1, \dots, K_m\}$ partition $K$ into nonempty groups. The merged law is

$$\mathrm{mrg}(\mu) := \sum_{j=1}^{m} w_j\, \delta_{s_j}, \qquad w_j := \sum_{k\in K_j} \lambda_k, \qquad s_j := \frac{1}{w_j} \sum_{k\in K_j} \lambda_k r_k.$$

Barycentres are preserved, so $\mathrm{mrg}(\mu) \in \mathcal{M}_q$. Let $T : K \to \Delta(\{1, \dots, m\})$ be the deterministic kernel with $T(j \mid k) = 1$ if and only if $k \in K_j$. Then

$$(C_\mu T)(j \mid a) = \sum_{k\in K_j} \frac{\lambda_k r_k(a)}{q(a)} = \frac{w_j s_j(a)}{q(a)} = C_{\mathrm{mrg}(\mu)}(j \mid a),$$

so clause (DP-rec) of Condition (P4) gives

$$C_\mu \succeq_q C_{\mathrm{mrg}(\mu)}.$$

Conversely, say that $\mu' \in \mathcal{M}_q$ is a *spread* of $\mu$ if $\mu'$ is obtained by replacing each atom $(\lambda_k, r_k)$ of $\mu$ by finitely many atoms $(\lambda_k w_{kj}, v_{kj})_j$ with $w_{kj} \geq 0$, $\sum_j w_{kj} = 1$, and $\sum_j w_{kj} v_{kj} = r_k$. Then $\mu = \mathrm{mrg}(\mu')$ for the grouping by $k$, so

$$C_{\mu'} \succeq_q C_\mu.$$

Atoms with $w_{kj} = 0$ are deleted from the list; and if the same posterior appears in several list entries, any two listings of the same measure induce the same posterior law at $q$, so the corresponding canonical channels are interchangeable by substitution. Merging records can only obscure which act was realised; splitting an atom into a barycentric family is the reverse operation.

*Step 2: bounded supports.* We claim the lemma holds under the additional hypothesis that all laws $\mu_{q,P_n}$ and $\mu_{q,Q_n}$ have support of size at most $K$, for some fixed $K$. Write

$$\mu_{q,P_n} = \sum_{k=1}^{K} \lambda_{n,k}\, \delta_{r_{n,k}},$$

padding with zero-mass atoms ($\lambda_{n,k} = 0$, $r_{n,k} := q$) so that exactly $K$ terms appear, and define the $K$-record channel $C_n : A \to \Delta(\{1, \dots, K\})$ by

$$C_n(k \mid a) := \frac{\lambda_{n,k}\, r_{n,k}(a)}{q(a)}.$$

Each row sums to one because $\sum_k \lambda_{n,k} r_{n,k} = q$, zero-mass columns contribute nothing, and $\mu_{q,C_n} = \mu_{q,P_n}$. Define $D_n$ from $\mu_{q,Q_n}$ in the same way. By substitution,

$$C_n \succeq_q D_n \qquad \text{for all } n.$$

The parameter sequences of $C_n$ and $D_n$ jointly live in the compact metric space $([0,1]\times\Delta(A))^{2K}$; pass to a single subsequence along which $\lambda_{n,k}\to\lambda_k^*$ and $r_{n,k}\to r_k^*$ for every $k$, simultaneously for both channels. Then $C_n\to C^*$ entrywise, where $C^*(k\mid a)=\lambda_k^* r_k^*(a)/q(a)$; each row of $C^*$ sums to one as an entrywise limit of rows summing to one, so $C^*$ is a channel. For every continuous $f$ on $\Delta(A)$,

$$\int f\,d\mu_{q,P_n}=\sum_{k=1}^K \lambda_{n,k} f(r_{n,k})\longrightarrow\sum_{k=1}^K \lambda_k^* f(r_k^*),$$

so by uniqueness of weak limits

$$\mu_{q,P}=\sum_{k=1}^K \lambda_k^*\,\delta_{r_k^*}=\mu_{q,C^*},$$

where the second equality holds because record $k$ of $C^*$ has probability $\lambda_k^*$ and, when $\lambda_k^*>0$, posterior $r_k^*$. Similarly $D_n\to D^*$ entrywise with $\mu_{q,D^*}=\mu_{q,Q}$. The closedness clause, Lemma 10(b), applied along the subsequence gives

$$C^*\succeq_q D^*,$$

and substitution returns $q^0\succeq_{P\sqcup Q} q^1$. This proves the bounded-support case; note that only the primitive closedness of Condition (P2) has been used, through Lemma 10.

*Step 3: the general case, by an $\varepsilon$-grid sandwich.* Write $\mu_n:=\mu_{q,P_n}$, $\nu_n:=\mu_{q,Q_n}$, $\mu:=\mu_{q,P}$, $\nu:=\mu_{q,Q}$. The only remaining gap is that the supports of $\mu_n,\nu_n$ need not be bounded. Fix $\varepsilon>0$. Two finite partitions of $\Delta(A)$ are used, one for each surrogate; they play independent roles and need not be related. For the spread, fix a finite triangulation $\mathcal{T}$ of $\Delta(A)$ into simplices of diameter at most $\varepsilon$, together with a measurable partition $\{\widetilde{S}:S\in\mathcal{T}\}$ of $\Delta(A)$ with $\widetilde{S}\subseteq S$, assigning each shared face to exactly one incident cell. For the merge, fix a finite Borel partition $\{E_1,\dots,E_M\}$ of $\Delta(A)$ into sets of diameter at most $\varepsilon$ whose relative boundaries carry no $\nu$-mass: cover $\Delta(A)$ by finitely many open balls of radius at most $\varepsilon/2$, choosing the radii so that no atom of $\nu$ lies on a bounding sphere ($\nu$ has finitely many atoms, so all but finitely many radii work), and disjointify.

For finite-support $\rho\in\mathcal{M}_q$ define two bounded-support surrogates. Both remain in $\mathcal{M}_q$: the spread operation replaces each posterior by a barycentric decomposition of itself, while the merge operation replaces a group of posteriors by their conditional mean, weighted by the total mass of the group. Cells with zero mass are ignored (or, equivalently, padded by an arbitrary posterior with zero weight when a fixed record alphabet is convenient). The *grid spread* $\mathrm{sp}_\varepsilon(\rho)$ replaces each atom $(\lambda,\beta)$ with $\beta\in\widetilde{S}$ by the atoms $\big(\lambda\,w_v(\beta),\,v\big)_v$, where $v$ ranges over the vertices of $S$ and $(w_v(\beta))_v$ are the barycentric coordinates of $\beta$ in $S$. It is a spread of $\rho$ in the sense of Step 1, is supported on the fixed vertex set of $\mathcal{T}$, and satisfies $C_{\mathrm{sp}_\varepsilon(\rho)}\succeq_q C_\rho$. The *cell merge* $\mathrm{mg}_\varepsilon(\rho)$ merges the atoms of $\rho$ within each class $E_j$ of the Borel partition; it has at most one atom per class and satisfies $C_\rho\succeq_q C_{\mathrm{mg}_\varepsilon(\rho)}$ by Step 1, since Step 1 allows any grouping of atoms.

By substitution and transitivity (Lemma 10), the hypothesis $P_n\succeq_q Q_n$ gives the chain

$$C_{\mathrm{sp}_\varepsilon(\mu_n)}\succeq_q C_{\mu_n}\succeq_q C_{\nu_n}\succeq_q C_{\mathrm{mg}_\varepsilon(\nu_n)}.$$

Both surrogate sequences converge weakly on fixed alphabets. For the spread side, let $f$ be continuous on $\Delta(A)$. The triangulation defines a continuous piecewise-affine interpolant $I_{\mathcal{T}}f$ by

$$I_{\mathcal{T}}f(r):=\sum_{v\in\mathrm{vert}(S)} w_v(r)f(v)\qquad\text{for } r\in S.$$

This is well-defined: on a shared face the barycentric coordinates computed in the two incident simplices put zero weight on vertices outside the face and coincide with the intrinsic barycentric coordinates on that face. Hence $I_{\mathcal{T}} f$ is continuous on all of $\Delta(A)$. Moreover, for every finite-support $\rho$,

$$\int f \, d\,\mathrm{sp}_\varepsilon(\rho) = \int I_{\mathcal{T}} f \, d\rho,$$

because each posterior is replaced by the barycentric lottery over the vertices of its simplex. Thus $\mu_n \Rightarrow \mu$ gives $\mathrm{sp}_\varepsilon(\mu_n) \Rightarrow \mathrm{sp}_\varepsilon(\mu)$. For the merge side, each class $E_j$ has relative boundary contained in the chosen bounding spheres, which carry no $\nu$-mass, so $E_j$ is a $\nu$-continuity set; therefore $\nu_n(E_j) \to \nu(E_j)$ and, since the truncated map $r \mapsto r\,\mathbf{1}_{E_j}(r)$ is bounded and its discontinuity set lies in the relative boundary of $E_j$, which is $\nu$-null, $\int_{E_j} r \, d\nu_n \to \int_{E_j} r \, d\nu$ for every class. Classes with $\nu(E_j) > 0$ thus have converging merged masses and merged locations. If $\nu(E_j) = 0$, then the merged mass tends to zero, so the chosen location in that class is immaterial for weak convergence; padding such a class by any fixed posterior only changes a zero-weight atom. Hence $\mathrm{mg}_\varepsilon(\nu_n) \Rightarrow \mathrm{mg}_\varepsilon(\nu)$.

The laws $\mathrm{sp}_\varepsilon(\mu_n)$ and $\mathrm{mg}_\varepsilon(\nu_n)$ have supports of size at most $N(\varepsilon)$, the maximum of the number of vertices of $\mathcal{T}$ and the number $M$ of classes of the Borel partition. Step 2 applied to the chain's endpoints therefore yields

$$C_{\mathrm{sp}_\varepsilon(\mu)} \succeq_q C_{\mathrm{mg}_\varepsilon(\nu)}.$$

Finally, send $\varepsilon \to 0$ along a sequence $\varepsilon_m \downarrow 0$ with triangulations and Borel partitions chosen as above. The law $\mathrm{sp}_{\varepsilon_m}(\mu)$ has at most $k_P\,|A|$ atoms, where $k_P$ is the number of atoms of $\mu$: each atom splits among the $|A|$ vertices of one cell. Each new atom lies within $\varepsilon_m$ of the atom it came from and group masses are preserved, so $\mathrm{sp}_{\varepsilon_m}(\mu) \Rightarrow \mu$. The law $\mathrm{mg}_{\varepsilon_m}(\nu)$ has at most $k_Q$ atoms, where $k_Q$ is the number of atoms of $\nu$: only classes meeting the support of $\nu$ carry mass. Once $\varepsilon_m$ is smaller than the minimal distance between distinct atoms of $\nu$, each class contains at most one atom of $\nu$ and $\mathrm{mg}_{\varepsilon_m}(\nu) = \nu$ exactly. The comparisons $C_{\mathrm{sp}_{\varepsilon_m}(\mu)} \succeq_q C_{\mathrm{mg}_{\varepsilon_m}(\nu)}$ hold for every $m$, the supports are uniformly bounded, and the laws converge weakly to $\mu$ and $\nu$; one further application of Step 2 gives $C_\mu \succeq_q C_\nu$. Substitution returns

$$q^0 \succeq_{P \sqcup Q} q^1. \qquad \square$$

The lemma shows that continuity of block comparisons in posterior-law convergence is not an independent behavioural commitment: it is the primitive closedness of Condition (P2) seen through the quotient that Condition (P3) and clause (DP-rec) of Condition (P4) impose, with the convex order supplying compactness where record alphabets grow without bound. Only weak preference is closed in the limit; no strictness claim is made or used.

**Stage 2: affine fibres and canonical boundary behaviour.** Public-coin independence and quotient continuity now permit Herstein–Milnor cardinalisation. The next lemmas record the finite integral form, normalize no information to zero, and make action coarsening, exact support restriction, and undetectable distinctions available before values are compared across priors. In the formal development these facts are packaged by a full-revelation-normalized selected representative. The paper now makes that same initial selection explicitly; a later coherent product gauge is declared when it changes the representative.

**Lemma 15** (Posterior-separable representation)**.** *Fix a full-support $q \in \Delta(A)$. There is a continuous affine functional*

$$F_q : \mathcal{M}_q \to \mathbb{R}$$

*such that, for any two channels $P, Q$ on $A$,*

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \Longleftrightarrow \quad F_q(\mu_{q,P}) \geq F_q(\mu_{q,Q}).$$

*Moreover, for some continuous $\phi_q : \Delta(A) \to \mathbb{R}$,*

$$F_q(\mu) = \int_{\Delta(A)} \phi_q(r)\, d\mu(r),$$

*and therefore*

$$F_q(\mu_{q,P}) = \sum_{x:m_{q,P}(x)>0} m_{q,P}(x)\, \phi_q(r_x^{q,P}).$$

*The affine functional is unique up to positive affine transformations. The integrand $\phi_q$ is not pointwise unique; adding an affine function of $r$ leaves its integral on the fibre $\mathcal{M}_q$ changed only by a constant.*

*Proof.* First note that every finite-support Bayes-plausible law is implemented by a finite channel. If

$$\mu = \sum_{k=1}^{n} \lambda_k \delta_{r_k} \in \mathcal{M}_q,$$

define

$$P(k \mid a) = \frac{\lambda_k r_k(a)}{q(a)}.$$

Since $q$ has full support and $\sum_k \lambda_k r_k = q$, this is a channel. Its record probability at $k$ is $\lambda_k$, and the posterior after $k$ is $r_k$; hence $\mu_{q,P} = \mu$.

Define a comparison on $\mathcal{M}_q$ as follows: for $\mu, \nu \in \mathcal{M}_q$, choose any channels $P, Q$ implementing them and compare the two block-supported copies of $q$ in $P \sqcup Q$. This comparison is independent of the chosen implementations. Indeed, if $P, P'$ both implement $\mu$ and $Q, Q'$ both implement $\nu$, then Lemma 12 gives

$$q^0 \sim_{P \sqcup P'} q^1, \qquad q^0 \sim_{Q \sqcup Q'} q^1.$$

Placing the relevant blocks in common block environments and using Lemma 10, transitivity gives

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \iff \quad q^0 \succeq_{P' \sqcup Q'} q^1.$$

Thus the quotient comparison on $\mathcal{M}_q$ is well defined; denote it by $\succeq_q$. This construction is only at the fixed full-support prior $q$, and every law in $\mathcal{M}_q$ is finite-support and, by the preceding construction, implemented by a finite channel. Lemma 10 also gives completeness and transitivity of this quotient comparison.

The set $\mathcal{M}_q$ is a mixture space under ordinary convex combination. It is convex because barycentres are preserved by convex combination. The quotient relation $\succeq_q$ is a weak order by Lemma 10. Public-coin mixtures of implementing channels correspond exactly to convex mixtures of posterior laws:

$$\mu_{q,\lambda P \oplus (1-\lambda) R} = \lambda \mu_{q,P} + (1-\lambda)\mu_{q,R}.$$

Therefore the derived public-coin independence property (Lemma 13) gives mixture independence: for every $\mu, \nu, \rho \in \mathcal{M}_q$ and $\lambda \in (0,1)$,

$$\mu \succeq_q \nu \quad \iff \quad \lambda\mu + (1-\lambda)\rho \succeq_q \lambda\nu + (1-\lambda)\rho.$$

The relative weak topology on $\mathcal{M}_q$ is inherited from the weak topology on probability measures over the compact metric simplex $\Delta(A)$, and convex-mixture paths are continuous in this topology. Lemma 14 gives relatively closed upper and lower contour sets. Indeed, if $\mu_n \Rightarrow \mu$ with $\mu_n, \mu, z \in \mathcal{M}_q$ and $\mu_n \succeq_q z$, choose finite implementing channels for these laws and apply Lemma 14 (with the constant sequence implementing $z$) to get $\mu \succeq_q z$; the argument for $\{\, \mu : z \succeq_q \mu \,\}$ is the same. Therefore, for fixed $\mu, \nu, \rho \in \mathcal{M}_q$, the sets

$$\{\lambda \in [0,1] : \lambda\mu + (1-\lambda)\nu \succeq_q \rho\} \quad \text{and} \quad \{\lambda \in [0,1] : \rho \succeq_q \lambda\mu + (1-\lambda)\nu\}$$

are closed.

By the Herstein–Milnor mixture-space theorem [3], there is a mixture-preserving representation $F_q : \mathcal{M}_q \to \mathbb{R}$:

$$\mu \succeq_q \nu \quad \iff \quad F_q(\mu) \geq F_q(\nu),$$

and

$$F_q(\lambda\mu + (1-\lambda)\nu) = \lambda F_q(\mu) + (1-\lambda)F_q(\nu).$$

If the quotient order is nonconstant, this representative is unique up to positive affine transformations. If the quotient order is constant, take $F_q \equiv 0$.

Moreover, $F_q$ is continuous in the relative weak topology. For any $z \in \mathcal{M}_q$, the sets

$$\{\mu : F_q(\mu) \geq F_q(z)\} = \{\mu : \mu \succeq_q z\} \quad \text{and} \quad \{\mu : F_q(\mu) \leq F_q(z)\} = \{\mu : z \succeq_q \mu\}$$

are relatively closed by the preceding Lemma 14 argument. Since $F_q$ is affine, its range on the convex set $\mathcal{M}_q$ is an interval. Hence every upper and lower level set of $F_q$ is relatively closed: levels outside the range are trivial, and levels inside the range equal $F_q(z)$ for some $z \in \mathcal{M}_q$. Thus $F_q$ is both upper and lower semicontinuous, and therefore continuous. For any channels $P, Q$ on $A$,

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \iff \quad F_q(\mu_{q,P}) \geq F_q(\mu_{q,Q}).$$

The no-information posterior law at $q$ is $\delta_q$. Write

$$\chi_q := \sum_{a \in A} q(a)\delta_{e_a}$$

for the full-revelation posterior law, where $e_a$ assigns probability one to action $a$.

It remains to write the affine functional in posterior-separable form, without leaving the finite-support domain. Let $X := \Delta(A)$. Choose

$$0 < \varepsilon < \min_{a \in A} q(a).$$

For each $r \in X$, define

$$s_r := \frac{q - \varepsilon r}{1 - \varepsilon} \in \Delta(A)$$

and

$$\eta_r := \varepsilon\delta_r + (1-\varepsilon)\sum_{a \in A} s_r(a)\delta_{e_a}.$$

Then $\eta_r \in \mathcal{M}_q$, because its barycentre is

$$\varepsilon r + (1-\varepsilon)s_r = q.$$

Choose constants $c_a$ such that

$$\sum_{a \in A} q(a)c_a = F_q(\chi_q),$$

for instance $c_a = F_q(\chi_q)$ for every $a$. Define

$$\phi_q(r) := \frac{F_q(\eta_r) - (1-\varepsilon)\sum_{a \in A} s_r(a)c_a}{\varepsilon}.$$

The map $r \mapsto \eta_r$ is weakly continuous and $F_q$ is continuous, so $\phi_q$ is continuous.

Now take any finite-support law

$$\mu = \sum_{k=1}^{n} \lambda_k \delta_{r_k} \in \mathcal{M}_q.$$

Since $\sum_k \lambda_k r_k = q$, we have

$$\sum_{k=1}^{n} \lambda_k s_{r_k} = q.$$

Therefore

$$\sum_{k=1}^{n} \lambda_k \eta_{r_k} = \varepsilon\mu + (1-\varepsilon)\chi_q.$$

By affinity of $F_q$,

$$\sum_{k=1}^{n} \lambda_k F_q(\eta_{r_k}) = F_q(\varepsilon\mu + (1-\varepsilon)\chi_q) = \varepsilon F_q(\mu) + (1-\varepsilon)F_q(\chi_q).$$

Using $\sum_k \lambda_k s_{r_k} = q$ and $\sum_a q(a)c_a = F_q(\chi_q)$ gives

$$\sum_{k=1}^{n} \lambda_k \phi_q(r_k) = F_q(\mu).$$

This is exactly

$$F_q(\mu) = \int_X \phi_q(r)\, d\mu(r)$$

for every finite-support $\mu \in \mathcal{M}_q$. Applying this to $\mu_{q,P}$ gives the stated finite-sum formula. The affine representative $F_q$ is unique up to positive affine transformations by Herstein–Milnor. The integrand is not pointwise unique on a fixed barycentre fibre: if $\ell$ is affine in $r$, then $\int \ell(r)\, d\mu(r) = \ell(q)$ for every $\mu \in \mathcal{M}_q$. □

**Lemma 16** (Refinement monotonicity and canonical normalisation)**.** *Fix a full-support $q \in \Delta(A)$ and let $F_q, \phi_q$ be as in Lemma 15. If a finite law $\mu \in \mathcal{M}_q$ is obtained from another finite law $\nu \in \mathcal{M}_q$ by replacing each posterior of $\nu$ by a finite Bayes-plausible continuation experiment, then*

$$F_q(\mu) \geq F_q(\nu).$$

*Consequently, $\phi_q$ may be chosen convex. After replacing $\phi_q$ by an observationally equivalent representative and subtracting a constant from $F_q$, one may impose*

$$\phi_q(q) = F_q(\delta_q) = 0, \qquad \phi_q(r) \geq 0 \quad \textit{for every } r \in \Delta(A),$$

*and therefore*

$$F_q(\mu) \geq 0 \quad \textit{for every } \mu \in \mathcal{M}_q.$$

*If $|A| \geq 2$, the representative can and will be selected uniquely by the additional full-revelation normalisation*

$$F_q(\chi_q) = 1.$$

*If $|A| = 1$, then $\mathcal{M}_q = \{\delta_q\}$ and we select $F_q \equiv 0$.*

*Proof.* Write

$$\nu = \sum_{i=1}^{n} m_i \delta_{r_i}, \qquad \mu = \sum_{i=1}^{n} m_i \nu_i, \qquad \nu_i \in \mathcal{M}_{r_i}.$$

Choose a channel $P : A \to \Delta(\{1, \ldots, n\})$ implementing $\nu$ at prior $q$. For each $i$, choose a finite continuation channel $Q^i : A \to \Delta(Z_i)$ implementing $\nu_i$ at prior $r_i$; on actions outside the support of $r_i$, define the rows arbitrarily. Those rows are never reached under prior $r_i$ and do not affect $\nu_i$; moreover, if $r_i(a) = 0$, then $q(a)P(i \mid a) = m_i r_i(a) = 0$, so they do not affect the compound posterior law at $q$. The compound channel

$$C := P \triangleright \{Q^i\}_{i=1}^{n}$$

implements $\mu$ at prior $q$. Let $T : \bigsqcup_{i=1}^{n}(\{i\} \times Z_i) \to \Delta(\{1,\dots,n\})$ be the kernel $T(i \mid (i,z)) = 1$; then $CT = P$. Clause (DP-rec) of Condition (P4) applied to $C$ and $T$ gives

$$q^0 \succeq_{C \sqcup P} q^1.$$

By Lemma 15,

$$F_q(\mu) \geq F_q(\nu).$$

To prove convexity, take $r, s_1, \dots, s_k \in \Delta(A)$ with

$$r = \sum_{j=1}^{k} \lambda_j s_j, \qquad \lambda_j \geq 0, \quad \sum_j \lambda_j = 1.$$

Choose $0 < \varepsilon < \min_a q(a)$ and set

$$t := \frac{q - \varepsilon r}{1 - \varepsilon}.$$

This belongs to $\Delta(A)$: for each $a$, $q(a) - \varepsilon r(a) \geq q(a) - \varepsilon > 0$, and the coordinates sum to one. The law

$$\varepsilon \sum_{j=1}^{k} \lambda_j \delta_{s_j} + (1-\varepsilon)\delta_t$$

is a refinement of

$$\varepsilon \delta_r + (1-\varepsilon)\delta_t,$$

and both belong to $\mathcal{M}_q$. Refinement monotonicity and the integral representation give

$$\varepsilon \sum_{j=1}^{k} \lambda_j \phi_q(s_j) + (1-\varepsilon)\phi_q(t) \geq \varepsilon \phi_q(r) + (1-\varepsilon)\phi_q(t).$$

Cancelling the common term and dividing by $\varepsilon$ yields

$$\phi_q(r) \leq \sum_{j=1}^{k} \lambda_j \phi_q(s_j),$$

so $\phi_q$ is convex.

Since $q$ is in the relative interior of the finite-dimensional simplex, the supporting-hyperplane theorem for finite convex functions gives a relative subgradient $g \in \mathbb{R}^A$ of the continuous convex function $\phi_q$ at $q$. Define

$$\widetilde{\phi}_q(r) := \phi_q(r) - g \cdot (r - q).$$

For every $\mu \in \mathcal{M}_q$,

$$\int g \cdot (r - q)\, d\mu(r) = g \cdot (b(\mu) - q) = 0,$$

so $\widetilde{\phi}_q$ represents the same affine functional on $\mathcal{M}_q$. The subgradient inequality gives

$$\widetilde{\phi}_q(r) \geq \widetilde{\phi}_q(q) \qquad \text{for all } r \in \Delta(A).$$

Finally replace

$$F_q(\mu) \quad \text{by} \quad F_q(\mu) - F_q(\delta_q)$$

and replace $\widetilde{\phi}_q$ by

$$\widetilde{\phi}_q - \widetilde{\phi}_q(q).$$

Before this final constant shift, the representation with $\widetilde{\phi}_q$ gives

$$F_q(\delta_q) = \int \widetilde{\phi}_q(r)\, d\delta_q(r) = \widetilde{\phi}_q(q).$$

Thus the same constant is subtracted from both sides of the integral representation, and subtracting a constant from $F_q$ preserves all ordinal comparisons on $\mathcal{M}_q$. The final normalisation gives

$$\phi_q(q) = F_q(\delta_q) = 0,$$

and makes $\phi_q$ non-negative on the whole simplex. Hence

$$F_q(\mu) = \int \phi_q(r)\, d\mu(r) \geq 0$$

for every $\mu \in \mathcal{M}_q$.

Suppose now that $|A| \geq 2$. Full revelation implements $\chi_q$ and no information implements $\delta_q$. The strict local-orientation clause of Condition (P1), transported through the quotient comparison, gives

$$F_q(\chi_q) > F_q(\delta_q) = 0.$$

Divide both $F_q$ and $\phi_q$ by the positive number $F_q(\chi_q)$. This preserves the integral representation, all comparisons, convexity, and non-negativity, and gives $F_q(\chi_q) = 1$. These two anchor values make the selected affine representative unique. If $|A| = 1$, every posterior law with barycentre $q$ is $\delta_q$, so the zero representative is the only relevant choice. $\square$

For a finitely supported posterior law $\mu = \sum_i \lambda_i \delta_{r_i}$ and a bijection $\pi : A \to A'$, write

$$\mu\pi := \sum_i \lambda_i \delta_{r_i \pi}$$

for the posterior law obtained by relabelling actions through $\pi$.

**Lemma 17** (Exact cross-fibre relabelling)**.** *Action coarsening implies ordinal cross-fibre equivalence for bijections. The zero/full-revelation selection in Lemma 16 makes that equivalence exact: for every bijection $\pi : A \to A'$ and full-support $q \in \Delta(A)$,*

$$F_{q\pi}(\mu\pi) = F_q(\mu)$$

*for all $\mu \in \mathcal{M}_q$. In particular, no-information experiments have the same selected value zero on every finite action set.*

*Proof. Ordinal equivalence.* Let $\pi : A \to A'$ be a bijection. View $\pi$ as the deterministic kernel $S(a' \mid a) = \mathbf{1}_{a'=\pi(a)}$. For any channel $P : A \to \Delta(X)$ and full-support $q \in \Delta(A)$, the Bayesian pushforward satisfies $S^q P = P\pi^{-1}$ (explicitly: $(S^q P)(x \mid a') = P(x \mid \pi^{-1}(a'))$). Clause (DP-act) of Condition (P4) gives

$$q^0 \succeq_{P \sqcup P\pi^{-1}} (q\pi)^1.$$

For the reverse inequality, apply (DP-act) to the bijection $\pi^{-1} : A' \to A$, the prior $q\pi \in \Delta(A')$, and the channel $P\pi^{-1} : A' \to \Delta(X)$. Then $(\pi^{-1})^{q\pi}(P\pi^{-1}) = P$ (the inverse bijection undoes the relabeling), and $(q\pi)\pi^{-1} = q$. Hence

$$(q\pi)^0 \succeq_{P\pi^{-1} \sqcup P} q^1.$$

By the irrelevant-blocks clause of Condition (P3), block comparisons depend only on the pair of channels involved, not on numeric labels. The comparison $(q\pi)^0 \succeq_{P\pi^{-1} \sqcup P} q^1$ asks whether $(q\pi, P\pi^{-1})$ is at least as good as $(q, P)$. Swapping the block labels (so that $P$ becomes block 0 and $P\pi^{-1}$ becomes block 1) gives the equivalent statement $q^0 \preceq_{P \sqcup P\pi^{-1}} (q\pi)^1$. Combining with the first inequality gives

$$q^0 \sim_{P \sqcup P\pi^{-1}} (q\pi)^1.$$

If $P$ implements $\mu \in \mathcal{M}_q$, then $P\pi^{-1}$ implements $\mu\pi \in \mathcal{M}_{q\pi}$: for every positive-probability record $x$,

$$m_{q\pi, P\pi^{-1}}(x) = m_{q,P}(x), \qquad r_x^{q\pi, P\pi^{-1}} = r_x^{q,P}\pi.$$

Now take arbitrary $\mu, \nu \in \mathcal{M}_q$, and choose channels $P, Q$ implementing them at prior $q$. Let $E$ be the four-block environment with blocks $P$, $P\pi^{-1}$, $Q$, and $Q\pi^{-1}$. Write $x, x', y, y'$ for the block-supported lotteries $q$, $q\pi$, $q$, and $q\pi$ on these four blocks, respectively. By Lemma 10, the two two-block indifferences give $x \sim_E x'$ and $y \sim_E y'$. Hence, by transitivity of the weak order on $E$,

$$x \succeq_E y \quad \Longleftrightarrow \quad x' \succeq_E y'.$$

Applying Lemma 10 again to delete irrelevant blocks gives

$$q^0 \succeq_{P \sqcup Q} q^1 \quad \Longleftrightarrow \quad (q\pi)^0 \succeq_{P\pi^{-1} \sqcup Q\pi^{-1}} (q\pi)^1.$$

By Lemma 15 on the two fibres, this is equivalent to

$$F_q(\mu) \geq F_q(\nu) \quad \Longleftrightarrow \quad F_{q\pi}(\mu\pi) \geq F_{q\pi}(\nu\pi).$$

The map $\mu \mapsto \mu\pi$ is an affine bijection from $\mathcal{M}_q$ to $\mathcal{M}_{q\pi}$, with inverse induced by $\pi^{-1}$. Therefore $G(\mu) := F_{q\pi}(\mu\pi)$ is a continuous affine representative of the same weak order on $\mathcal{M}_q$ as $F_q$. Herstein–Milnor uniqueness gives $G = aF_q + b$ with $a > 0$. Since $\delta_q\pi = \delta_{q\pi}$ and Lemma 16 gives

$$F_q(\delta_q) = F_{q\pi}(\delta_{q\pi}) = 0,$$

we have $b = 0$. If $|A| = 1$, both representatives vanish identically. If $|A| \geq 2$, relabelling sends full revelation to full revelation, $\chi_q\pi = \chi_{q\pi}$, while the canonical selection gives $F_q(\chi_q) = F_{q\pi}(\chi_{q\pi}) = 1$. Substitution in $G = aF_q$ therefore gives $a = 1$. Hence

$$F_{q\pi}(\mu\pi) = F_q(\mu)$$

for all $\mu \in \mathcal{M}_q$.

*No-information equivalence.* By Lemma 16, $F_q(\delta_q) = 0$ for every full-support $q$ on every finite action set, independently of cardinality. □

**Lemma 18** (Undetectable distinctions neutrality)**.** *Let $S : A \to A'$ be an onto deterministic partition map, and let $P : A \to \Delta(X)$ be $S$-measurable:*

$$S(a) = S(b) \quad \Rightarrow \quad P(\cdot \mid a) = P(\cdot \mid b).$$

*Then for every full-support $q \in \Delta(A)$,*

$$q^0 \sim_{P \sqcup S^q P} (qS)^1.$$

*In particular, if $G_S : A \to \Delta(A')$ reveals the block $S(a)$, then*

$$(q, G_S) \sim (qS, \mathrm{Id}_{A'}).$$

*Proof.* Since $S$ is onto and $q$ has full support, $qS$ has full support on $A'$. Thus the Bayesian pushforward $S^q P$ has no zero-row ambiguity in this lemma. View the deterministic map $S$ as its associated stochastic kernel. One direction is exactly clause (DP-act) of Condition (P4):

$$q^0 \succeq_{P \sqcup S^q P} (qS)^1.$$

For the reverse direction, define the Bayesian refinement kernel $R : A' \to \Delta(A)$ by

$$R(a \mid a') = \begin{cases} q(a)/(qS)(a'), & S(a) = a', \\ 0, & \text{otherwise.} \end{cases}$$

For each $a' \in A'$, $R(\cdot \mid a')$ is stochastic because

$$\sum_{a \in A} R(a \mid a') = \frac{\sum_{a:S(a)=a'} q(a)}{(qS)(a')} = 1.$$

Moreover, for every $a \in A$,

$$((qS)R)(a) = \sum_{a' \in A'} (qS)(a')R(a \mid a') = (qS)(S(a))\frac{q(a)}{(qS)(S(a))} = q(a) > 0.$$

Thus $(qS)R = q$.

Since $P$ is $S$-measurable, the row of $S^q P$ at each block $a'$ equals the common $P$-row on that block: if $S(a) = a'$, then

$$S^q P(x \mid a') = \frac{\sum_{b:S(b)=a'} q(b)P(x \mid b)}{(qS)(a')} = P(x \mid a).$$

Now pushing $S^q P$ back through $R$ recovers $P$. For each $a \in A$,

$$(R^{qS}(S^q P))(x \mid a) = \frac{\sum_{a' \in A'} (qS)(a')R(a \mid a')S^q P(x \mid a')}{((qS)R)(a)} = S^q P(x \mid S(a)) = P(x \mid a).$$

Therefore

$$R^{qS}(S^q P) = P.$$

Apply (DP-act) to $(qS, S^q P)$ with the refinement kernel $R$:

$$(qS)^0 \succeq_{S^q P \sqcup P} q^1.$$

By Condition (P3) applied to the two-block environment $P \sqcup S^q P$ with ordered pair $(1, 0)$, this is equivalent to

$$(qS)^1 \succeq_{P \sqcup S^q P} q^0.$$

Together the two inequalities give the desired indifference. $\square$

For a nonempty $B \subseteq A$ and a finite law $\mu$ whose barycentre is supported on $B$, non-negativity implies that every atom $p$ of $\mu$ is supported on $B$. Define its support projection by

$$\pi_B(\mu) := \sum_{p \in \mathrm{supp}(\mu)} \mu(\{p\})\, \delta_{p|_B}.$$

It is a finite law on $\Delta(B)$ with barycentre $b(\mu)|_B$.

**Lemma 19** (Support restriction)**.** *Let $r \in \Delta(A)$ have support $B := \mathrm{supp}(r) \subsetneq A$. Then:*

*(i)* Posterior-law reduction. *For any channel $P : A \to \Delta(X)$, the posterior law $\mu_{r,P}$ depends only on the rows $P(\cdot \mid b)$ for $b \in B$. Rows $P(\cdot \mid a)$ with $a \notin B$ have zero contribution because $r(a) = 0$.*

*(ii)* Preference equivalence. *Two channels $P, Q : A \to \Delta(X)$ that agree on $B$ (i.e. $P(\cdot \mid b) = Q(\cdot \mid b)$ for all $b \in B$) are indifferent at prior $r$.*

*(iii)* Face identification. *For every channel $P : A \to \Delta(X)$,*

$$(r, P) \sim (r|_B, P|_B)$$

*as a two-block comparison; consequently the comparison between channels $P, Q$ at prior $r$ agrees with the comparison between $P|_B, Q|_B$ at the full-support prior $r|_B \in \Delta(B)$. This is proved by projecting $A$ to $B$ and embedding $B$ back into $A$, using the all-completions form of clause* (DP-act) *of Condition* (P4) *for the embedding.*

*(iv)* Representative. *Define $F_r : \mathcal{M}_r \to \mathbb{R}$ by $F_r(\mu) := F^B_{r|_B}(\pi_B(\mu))$, where $\pi_B$ projects posterior laws from $\Delta(A)$ to $\Delta(B)$ and $F^B_{r|_B}$ is the Lemma 15 representative on $\Delta(B)$. This $F_r$ represents continuation comparisons at prior $r$ in action set $A$.*

*Proof.* In this proof, $(s, R) \succeq (t, T)$ abbreviates the block comparison $s^0 \succeq_{R \sqcup T} t^1$, and $\sim$ denotes both weak inequalities. These comparisons involve different priors and different action sets; they chain by transitivity in Lemma 10(a), which is stated across such pairs.

Part (i): The marginal probability $m_{r,P}(x) = \sum_a r(a)P(x \mid a)$ and the Bayes numerator $r(a)P(x \mid a)$ are both zero for $a \notin B$. Hence $\mu_{r,P}$ is determined by $\{P(\cdot \mid b) : b \in B\}$.

Part (ii): Let $P$ and $\widetilde{P}$ be channels on $A$ that agree on $B$. We first show $(r, P) \sim (r, \widetilde{P})$. Fix $b_0 \in B$, and let $S : A \to B$ be the deterministic map with $S(b) = b$ for $b \in B$ and $S(a) = b_0$ for $a \notin B$. Then $rS = r|_B$ and the Bayesian pushforward $S^r P$ is $P|_B$ on $B$. Clause (DP-act) gives

$$(r, P) \succeq (r|_B, P|_B).$$

Let $I : B \hookrightarrow A$ be the inclusion. Then $(r|_B)I = r$, and the Bayesian pushforward $I^{r|_B}(P|_B)$ is pinned down on $B$ and unconstrained on $A \setminus B$. By the all-completions convention in (DP-act), choose the completion to be $\widetilde{P}$. Clause (DP-act) gives

$$(r|_B, P|_B) \succeq (r, \widetilde{P}).$$

Thus $(r, P) \succeq (r, \widetilde{P})$. Applying the same projection/inclusion argument with $\widetilde{P}$ as the original channel and choosing $P$ as the inclusion completion gives $(r, \widetilde{P}) \succeq (r, P)$. Hence $(r, P) \sim (r, \widetilde{P})$.

If $P, Q$ agree on $B$, choose a common extension $\widetilde{P} = \widetilde{Q}$ of this common restriction to all of $A$ (for instance, use one fixed off-support row). The preceding paragraph gives $(r, P) \sim (r, \widetilde{P})$ and $(r, Q) \sim (r, \widetilde{Q})$; since $\widetilde{P} = \widetilde{Q}$, common-environment transitivity gives $(r, P) \sim (r, Q)$.

Part (iii): We first prove $(r, P) \sim (r|_B, P|_B)$ for every channel $P : A \to \Delta(X)$.

*Forward direction (*(DP-act) *projection).* Let $S : A \to B$ map all $a \notin B$ to some fixed $b_0 \in B$. The coarsened prior is $rS = r|_B$ (since $r(a) = 0$ for $a \notin B$), and the Bayesian pushforward $S^r P$ equals $P|_B$ on $B$. By (DP-act),

$$(r, P) \succeq (r|_B, P|_B).$$

*Reverse direction (*(DP-act) *inclusion with completion).* Let $I : B \hookrightarrow A$ be the inclusion. Then $(r|_B)I = r$. The Bayesian pushforward $I^{r|_B}(P|_B)$ is pinned down on $B$ and arbitrary on $A \setminus B$. Choose the (DP-act) completion to be any extension $\widetilde{P} : A \to \Delta(X)$ of $P|_B$. Then

$$(r|_B, P|_B) \succeq (r, \widetilde{P}).$$

By Part (ii), $(r, \widetilde{P}) \sim (r, P)$ because $\widetilde{P}$ and $P$ agree on $B = \mathrm{supp}(r)$. Hence $(r|_B, P|_B) \succeq (r, P)$.

Together the two directions give $(r, P) \sim (r|_B, P|_B)$.

*Comparison equivalence.* For any $P, Q$: $(r, P) \succeq (r, Q)$ iff (by the above indifferences $(r, P) \sim (r|_B, P|_B)$, $(r, Q) \sim (r|_B, Q|_B)$, and Condition (P1) transitivity) $(r|_B, P|_B) \succeq (r|_B, Q|_B)$.

Part (iv): If $\mu \in \mathcal{M}_r$, every posterior in the finite support of $\mu$ is supported on $B$; otherwise the barycentre $r$ would assign positive mass to $A \setminus B$. Hence $\pi_B(\mu) \in \mathcal{M}_{r|_B}$ is well defined. By (iii), continuation comparisons at $r$ are isomorphic to those at $r|_B$. The Lemma 15 representative $F^B_{r|_B}$ on $\Delta(B)$ therefore induces a representative on $\mathcal{M}_r$ via $\pi_B$. □

**Stage 3: product transport and the cross-prior bridge.** An independent background is first shown to be ordinally inert. Separate affinity then yields the most general compatible product expression, $x + y + \kappa xy$, after a coherent positive gauge. Crucially, Lemma 23 is proved before any coarse-revelation value is used: product structure and face scales alone do not identify cross-prior block comparisons.

**Lemma 20** (Derived background inertness)**.** *Assume Conditions* (P1) *and* (P3)–(P5)*. Let $A_1, A_2$ be finite action sets and let $q_1 \in \Delta(A_1)$ and $q_2 \in \Delta(A_2)$ have full support. Then, for all channels $P_1, Q_1$ on $A_1$ and every background channel $R_2$ on $A_2$,*

$$(q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes R_2) \sqcup (Q_1 \otimes R_2)} (q_1 \otimes q_2)^1 \quad \Longleftrightarrow \quad q_1^0 \succeq_{P_1 \sqcup Q_1} q_1^1, \tag{BI}$$

*and symmetrically for comparisons in the second component with a common first-component background. In particular the left-hand comparison does not depend on the common background: for all backgrounds $R_2, S_2$ on $A_2$,*

$$(q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes R_2) \sqcup (Q_1 \otimes R_2)} (q_1 \otimes q_2)^1 \quad \Longleftrightarrow \quad (q_1 \otimes q_2)^0 \succeq_{(P_1 \otimes S_2) \sqcup (Q_1 \otimes S_2)} (q_1 \otimes q_2)^1.$$

*Proof.* Throughout, for a prior $s$ on an action set $A$ and channels $V, W$ on $A$ write $V \succeq_s W$ for the block comparison $s^0 \succeq_{V \sqcup W} s^1$, and $(s, V) \sim (t, W)$ for the two-block indifference between the pairs. By Lemma 10(a) these comparisons form a weak order for each fixed prior and chain transitively across pairs with different priors and action sets. By Lemma 12, at a full-support prior a channel may be replaced by any channel with the same posterior law, in weak and in strict comparisons alike; we call this *substitution.*

*Preliminary reduction: a common foreground alphabet.* Condition (P5) requires the two continuation channels used on a given branch to share a record alphabet. If $P_1 : A_1 \to \Delta(X)$ and $Q_1 : A_1 \to \Delta(X')$, replace them by their zero-padded versions on $X \sqcup X'$, as in Step 2 of Lemma 13. Padding adds only null records, so it changes neither $\mu_{q_1,\cdot}$ nor $\mu_{q_1 \otimes q_2, \cdot \otimes R_2}$; both sides of (BI) are therefore unaffected by substitution at the full-support priors $q_1$ and $q_1 \otimes q_2$. Assume from now on that $P_1$ and $Q_1$ share the record alphabet $X_1$.

For a channel $V : A_1 \to \Delta(X_1)$ and a finite set $A_2$, write $\widetilde{V}$ for its lift to the product action set,

$$\widetilde{V}(z \mid (a_1, a_2)) := V(z \mid a_1),$$

and let $\widetilde{R}_2(x_2 \mid (a_1, a_2)) := R_2(x_2 \mid a_2)$ be the lift of the background.

*Step 1: a product channel is a two-stage channel with background first stage.* Take $\widetilde{R}_2$ as first stage on the action set $A_1 \times A_2$ and the constant continuation profile $\widetilde{V}$ in every branch, so that every branch uses the same continuation alphabet $X_1$ and the compound record set $\bigsqcup_{x_2 \in X_2} (\{x_2\} \times X_1)$ is $X_2 \times X_1$. By the definition of two-stage composition,

$$(\widetilde{R}_2 \triangleright \{\widetilde{V}\}_{x_2})(x_2, z \mid (a_1, a_2)) = R_2(x_2 \mid a_2)\, V(z \mid a_1) = (V \otimes R_2)(z, x_2 \mid (a_1, a_2)).$$

The two channels therefore differ only by the bijection $(x_2, z) \mapsto (z, x_2)$ of record labels, which leaves every record probability and every posterior unchanged. Hence they induce the same posterior law at $q_1 \otimes q_2$, and substitution makes them interchangeable in every comparison at that prior. Applying this to $V = P_1$ and $V = Q_1$, the left-hand side of (BI) is equivalent to

$$\widetilde{R}_2 \triangleright \{\widetilde{P}_1\} \ \succeq_{q_1 \otimes q_2} \ \widetilde{R}_2 \triangleright \{\widetilde{Q}_1\}, \tag{$*$}$$

and likewise for the strict comparisons.

*Step 2: every reached branch posterior is a product with first marginal $q_1$.* Write $q := q_1 \otimes q_2$, $m(x_2) := m_{q, \widetilde{R}_2}(x_2)$ and, for $m(x_2) > 0$, $r_{x_2} := r_{x_2}^{q, \widetilde{R}_2}$. Since the rows of $\widetilde{R}_2$ do not depend on $a_1$,

$$m(x_2) = \sum_{a_2 \in A_2} q_2(a_2) R_2(x_2 \mid a_2) = m_{q_2, R_2}(x_2),$$

and Bayes' rule gives

$$r_{x_2}(a_1, a_2) = \frac{q_1(a_1) q_2(a_2) R_2(x_2 \mid a_2)}{m(x_2)} = q_1(a_1)\, q_2^{x_2}(a_2), \qquad q_2^{x_2} := r_{x_2}^{q_2, R_2}.$$

Thus $r_{x_2} = q_1 \otimes q_2^{x_2}$: the background signal may change beliefs about $a_2$, but in every reached branch the two components stay independent and the first marginal is still $q_1$. At least one branch has positive probability.

*Step 3: a lifted foreground channel is evaluated as on its own component.* We claim that for every $s_2 \in \Delta(A_2)$ and every channel $V : A_1 \to \Delta(X_1)$,

$$(q_1 \otimes s_2, \widetilde{V}) \sim (q_1, V). \tag{†}$$

First suppose $s_2$ has full support, so that $q_1 \otimes s_2$ has full support on $A_1 \times A_2$. Let $S : A_1 \times A_2 \to A_1$ be the coordinate projection, an onto deterministic map, and note that $\widetilde{V}$ is $S$-measurable, since $S(a) = S(b)$ implies $\widetilde{V}(\cdot \mid a) = \widetilde{V}(\cdot \mid b)$. Moreover $(q_1 \otimes s_2)S = q_1$ and, for every $a_1$,

$$(S^{q_1 \otimes s_2} \widetilde{V})(z \mid a_1) = \frac{\sum_{a_2 \in A_2} q_1(a_1) s_2(a_2) V(z \mid a_1)}{q_1(a_1)} = V(z \mid a_1),$$

so the Bayesian pushforward of $\widetilde{V}$ under $S$ is $V$ itself. Undetectable-distinctions neutrality (Lemma 18) gives (†).

If $s_2$ is not full support, let $B_2 := \operatorname{supp}(s_2)$, so that $B := A_1 \times B_2$ is the support of $q_1 \otimes s_2$ and $B \subsetneq A_1 \times A_2$. The set $B$ *is* the product action set $A_1 \times B_2$, so no relabelling is involved: the restricted prior is $q_1 \otimes (s_2|_{B_2})$ and the restricted channel $\widetilde{V}|_B$ is the lift of $V$ to $A_1 \times B_2$. Face identification (Lemma 19(iii)) gives $(q_1 \otimes s_2, \widetilde{V}) \sim (q_1 \otimes s_2|_{B_2}, \widetilde{V}|_B)$, and the full-support case just proved, applied on the action set $A_1 \times B_2$, gives $(q_1 \otimes s_2|_{B_2}, \widetilde{V}|_B) \sim (q_1, V)$. Transitivity in a common block environment yields (†) in general.

Fix a branch with $m(x_2) > 0$ and apply (†) at $s_2 = q_2^{x_2}$ to $V = P_1$ and to $V = Q_1$, using $r_{x_2} = q_1 \otimes q_2^{x_2}$ from Step 2. Place the four pairs $(r_{x_2}, \widetilde{P}_1)$, $(q_1, P_1)$, $(r_{x_2}, \widetilde{Q}_1)$, $(q_1, Q_1)$ as blocks of one finite block environment (Condition (P3), Lemma 10). The two indifferences and transitivity of the weak order in that environment (Condition (P1)) give

$$\widetilde{P}_1 \succeq_{r_{x_2}} \widetilde{Q}_1 \quad \Longleftrightarrow \quad P_1 \succeq_{q_1} Q_1, \tag{‡}$$

and, since the relation is complete, the same equivalence for the strict comparisons. In particular the branch comparison is the same in every reached branch and does not involve $R_2$.

*Step 4: forward implication.* Assume $P_1 \succeq_{q_1} Q_1$. By (‡), the branch hypothesis $r_{x_2}^0 \succeq_{\widetilde{P}_1 \sqcup \widetilde{Q}_1} r_{x_2}^1$ of Condition (P5) holds for every branch with $m(x_2) > 0$, with the common first stage $\widetilde{R}_2$ and the two constant continuation profiles. Condition (P5) yields ($*$), hence the left-hand side of (BI) by Step 1.

*Step 5: converse, via the strictness clause.* Assume the left-hand side of (BI), so that ($*$) holds by Step 1, and suppose $P_1 \succeq_{q_1} Q_1$ fails. By completeness and the reverse-block equivalence of Lemma 10, $Q_1 \succ_{q_1} P_1$, so by the strict form of (‡) every reached branch satisfies $\widetilde{Q}_1 \succ_{r_{x_2}} \widetilde{P}_1$. At least one such branch exists and has positive probability, so the strictness clause of Condition (P5), applied with the two profiles exchanged, gives

$$\widetilde{R}_2 \triangleright \{\widetilde{Q}_1\} \succ_{q_1 \otimes q_2} \widetilde{R}_2 \triangleright \{\widetilde{P}_1\},$$

contradicting ($*$) because $\succeq_{q_1 \otimes q_2}$ is a weak order (Lemma 10). Hence $P_1 \succeq_{q_1} Q_1$, proving (BI).

The second-component clause is the same argument with the roles of the two coordinates exchanged: the common first-component background is used as the first stage and the second-component channels as the continuations. Finally, since both product comparisons in the last display of the lemma are equivalent to the single background-free comparison $P_1 \succeq_{q_1} Q_1$, they are equivalent to each other. □

The lemma shows that separability across independent components is not an independent behavioural commitment. A product channel is a two-stage channel whose first stage is the background experiment: it can teach the observer about the background action, but it leaves the foreground component independent of it and its marginal unchanged, so in every reached branch the continuation comparison is the foreground comparison itself. Branchwise monotonicity then transmits that single comparison to the compound, and its strictness clause transmits it back. The derived statement is stronger than mere invariance to changing the background: it identifies the product comparison with the comparison of the foreground channels alone, which is the form used in Lemma 21 below.

For finite posterior laws $\mu = \sum_i m_i \delta_{r_i} \in \mathsf{X}_A$ and $\nu = \sum_j n_j \delta_{s_j} \in \mathsf{X}_B$, define

$$\mu \otimes \nu := \sum_{i,j} m_i n_j \, \delta_{r_i \otimes s_j}.$$

Bayes' rule then gives the product factorisation

$$\mu_{q_1 \otimes q_2, \, P_1 \otimes P_2} = \mu_{q_1, P_1} \otimes \mu_{q_2, P_2}. \tag{PL}$$

**Lemma 21** (Coherent product quasi-additivity)**.** *There is a choice of positive rescalings of the zero-normalised fibrewise representatives $F_q$, still denoted $F_q$, and a scalar $\kappa \in \mathbb{R}$ such that, for all full-support priors $p \in \Delta(A)$ and $r \in \Delta(B)$ and all $\mu \in \mathcal{M}_p$, $\nu \in \mathcal{M}_r$,*

$$F_{p \otimes r}(\mu \otimes \nu) = F_p(\mu) + F_r(\nu) + \kappa F_p(\mu) F_r(\nu). \tag{QA}$$

*If one factor is a singleton, the same formula holds with its identically zero representative. Moreover, for every nondegenerate $r$ and every $\nu \in \mathcal{M}_r$,*

$$1 + \kappa F_r(\nu) > 0. \tag{POS}$$

*Proof.* All representatives in this proof are zero-normalised: $F_q(\delta_q) = 0$. The proof first treats nondegenerate action sets, meaning action sets with at least two elements.

*Step 1: product-coordinate independence.* Fix full-support priors $p$ and $r$ on nondegenerate finite action sets and set

$$L_{p,r}(\mu, \nu) := F_{p \otimes r}(\mu \otimes \nu).$$

This map is continuous and separately affine. To identify a first-coordinate slice, let $P, Q$ implement $\mu, \mu' \in \mathcal{M}_p$ and let $R$ implement $\nu \in \mathcal{M}_r$. By the product factorisation (PL), $P \otimes R$ implements $\mu \otimes \nu$ and $Q \otimes R$ implements $\mu' \otimes \nu$ at the full-support prior $p \otimes r$. Hence Lemma 15, applied on the fibre $\mathcal{M}_{p \otimes r}$ and on the fibre $\mathcal{M}_p$, turns background inertness (BI) of Lemma 20 into

$$L_{p,r}(\mu, \nu) \geq L_{p,r}(\mu', \nu) \quad \iff \quad F_p(\mu) \geq F_p(\mu').$$

The symmetric clause of Lemma 20 identifies every second-coordinate slice with the $F_r$-order. Local non-triviality, the fibrewise representation, and the zero normalisation from Lemma 16 make both slice orders nonconstant: $F_p(\chi_p) > 0$ and $F_r(\chi_r) > 0$.

*Step 2: the pair-specific bilinear form.* Put $x(\mu) := F_p(\mu)$ and $y(\nu) := F_r(\nu)$. For fixed $\nu$, affine-utility uniqueness gives

$$L_{p,r}(\mu, \nu) = \alpha(\nu) x(\mu) + \gamma(\nu), \qquad \alpha(\nu) > 0.$$

Taking $\mu = \delta_p$ shows $\gamma(\nu) = L_{p,r}(\delta_p, \nu)$. This is a continuous affine representative of the $F_r$-order and vanishes at $\delta_r$, so

$$\gamma(\nu) = B_{p,r} y(\nu)$$

for some $B_{p,r} > 0$. Likewise, the slice at $\nu = \delta_r$ gives

$$L_{p,r}(\mu, \delta_r) = A_{p,r} x(\mu)$$

for some $A_{p,r} > 0$.

Let $\bar{\mu} := \chi_p$ and $h_p := F_p(\chi_p) > 0$. The function $\nu \mapsto L_{p,r}(\bar{\mu}, \nu)$ is again a continuous affine representative of the $F_r$-order. Hence there are $D_{p,r} > 0$ and a constant $E_{p,r}$ such that

$$L_{p,r}(\bar{\mu}, \nu) = D_{p,r} y(\nu) + E_{p,r}.$$

At $\nu = \delta_r$, $E_{p,r} = A_{p,r} h_p$. Comparing this identity with $L_{p,r}(\bar{\mu}, \nu) = \alpha(\nu) h_p + B_{p,r} y(\nu)$ gives

$$\alpha(\nu) = A_{p,r} + C_{p,r} y(\nu), \qquad C_{p,r} := \frac{D_{p,r} - B_{p,r}}{h_p}.$$

Therefore

$$L_{p,r}(\mu, \nu) = A_{p,r} x(\mu) + B_{p,r} y(\nu) + C_{p,r} x(\mu) y(\nu). \tag{BIL}$$

The coordinate-slice slopes are positive:

$$A_{p,r} + C_{p,r} y(\nu) > 0, \qquad B_{p,r} + C_{p,r} x(\mu) > 0.$$

*Step 3: normalising the two linear coefficients.* Let

$$\alpha : ((A \times B) \times C) \longrightarrow A \times (B \times C), \qquad \alpha((a, b), c) = (a, (b, c)),$$

be the canonical associating bijection. It carries the left-grouped product prior and posterior law to the right-grouped ones. Exact cross-fibre relabelling (Lemma 17) therefore makes the two evaluations equal, with no unknown affine factor. Expanding this common value under the two groupings and comparing the coefficients of $x$, $y$, and $z$ gives

$$A_{p\otimes r,s} A_{p,r} = A_{p,r\otimes s}, \tag{C1}$$
$$A_{p\otimes r,s} B_{p,r} = B_{p,r\otimes s} A_{r,s}, \tag{C2}$$
$$B_{p\otimes r,s} = B_{p,r\otimes s} B_{r,s}. \tag{C3}$$

Coefficient comparison is legitimate because the mixtures $(1-t)\delta_p + t\chi_p$ make $x$ range over a nondegenerate interval, and similarly in the other coordinates.

The coordinate swap is another bijection, so exact cross-fibre relabelling gives

$$F_{r\otimes p}(\nu \otimes \mu) = F_{p\otimes r}(\mu \otimes \nu).$$

Comparing coefficients in (BIL),

$$B_{r,p} = A_{p,r}, \qquad A_{r,p} = B_{p,r}, \qquad C_{r,p} = C_{p,r}.$$

Thus, with $\rho(p, r) := B_{p,r}/A_{p,r}$,

$$\rho(r, p) = \rho(p, r)^{-1}. \tag{SW}$$

Divide (C2) by (C1):

$$\rho(p, r) = \rho(p, r \otimes s) A_{r,s}.$$

Fix a nondegenerate reference prior $q_0$ on a finite action set $A_0$ and put $g(r) := \rho(q_0, r)$. Then

$$A_{r,s} = \frac{g(r)}{g(r \otimes s)}. \tag{G}$$

The coefficients $A_{p,r}, B_{p,r}, C_{p,r}$ are unchanged when either prior is transported by a bijection. Indeed, transport the product law by the product bijection, use Lemma 17 on the two input fibres and the product fibre, and compare the unique coefficients in (BIL). Hence $\rho$, and therefore $g$, is invariant under relabelling. In particular it takes the same value on the two associations of a product prior.

Rescale every representative by $F_p^* := g(p)F_p$. Under this positive rescaling,

$$A^*_{p,r} = A_{p,r}\frac{g(p\otimes r)}{g(p)}, \quad B^*_{p,r} = B_{p,r}\frac{g(p\otimes r)}{g(r)}, \quad C^*_{p,r} = C_{p,r}\frac{g(p\otimes r)}{g(p)g(r)}.$$

Because $g$ is relabelling-invariant, these rescaled representatives retain the exact relabelling equality of Lemma 17. Thus the associator and swap comparisons used below still carry no scalar defect. Equation (G) gives $A^*_{p,r} = 1$ for every pair. The transformed swap identity gives $B^*_{p,r} = A^*_{r,p} = 1$. Thus, after the positive gauge rescaling, $A_{p,r} = B_{p,r} = 1$ for every nondegenerate pair (we now drop the stars). Positivity of the slice slopes is preserved because the rescaling factors are positive.

For a boundary prior $r \in \Delta(A)$ with $B = \operatorname{supp}(r)$, redefine the post-gauge boundary representative by exact support transport,

$$F_r^A(\mu) := F^B_{r|_B}(\pi_B(\mu)),$$

where the representative on the right is the newly gauged full-support one; singleton support faces remain identically zero. Thus the support identity of Lemma 19 remains definitional after the gauge change.

*Step 4: the interaction coefficient is universal.* Write the remaining coefficient as $\kappa_{p,r}$. Comparing the $xy$, $xz$, $yz$, and $xyz$ coefficients in the two associative expansions gives

$$\kappa_{p,r} = \kappa_{p,r\otimes s}, \tag{K1}$$

$$\kappa_{p\otimes r,s} = \kappa_{p,r\otimes s}, \tag{K2}$$

$$\kappa_{p\otimes r,s} = \kappa_{r,s}, \tag{K3}$$

$$\kappa_{p\otimes r,s}\kappa_{p,r} = \kappa_{p,r\otimes s}\kappa_{r,s}. \tag{K4}$$

The last equation follows from the first three, but it records the full coefficient comparison. After $A = B = 1$, exact invariance under the coordinate swap and comparison of the bilinear coefficient give

$$\kappa_{p,r} = \kappa_{r,p}. \tag{KS}$$

Using (K1)–(K3) with the reference prior $q_0$,

$$\kappa_{p,r} = \kappa_{p,r\otimes q_0} = \kappa_{p\otimes r,q_0} = \kappa_{r,q_0}.$$

Hence $\kappa_{p,r}$ is independent of its first argument. By symmetry it is also independent of its second argument. Denote the common value by $\kappa$. This proves (QA) for nondegenerate factors.

The positive slice-slope condition now reads

$$1 + \kappa F_r(\nu) > 0,$$

which is (POS).

*Step 5: singleton factors.* Let $q_* = \delta_*$ be the prior on a singleton. Its zero-normalised representative is identically zero. If $p$ is nondegenerate, the canonical bijection $A \to A\times\{*\}$ and exact relabeling give

$$F_{p\otimes q_*}(\mu\otimes\delta_{q_*}) = F_p(\mu).$$

The product gauge preserves this equality because $p\otimes q_*$ is a relabelling of $p$. Since $F_{q_*} = 0$, equation (QA) follows immediately. The reversed order is identical, and the two-singleton case is $0 = 0$. □

**Corollary 22** (Exact relabelling after the product gauge)**.** *For every bijection $\pi : A \to A'$ and every full-support $q \in \Delta(A)$,*

$$F_{q\pi}(\mu\pi) = F_q(\mu) \qquad (\mu \in \mathcal{M}_q).$$

*Proof.* Before the product gauge this is Lemma 17. The proof of Lemma 21 shows that the gauge factor $g(q)$ is unchanged by every bijection, so multiplication by that factor preserves the equality. The singleton case is immediate because both representatives vanish. $\square$

**Lemma 23** (Cross-prior block representation)**.** *For arbitrary finite priors $q \in \Delta(A)$ and $r \in \Delta(B)$ and channels $P : A \to \Delta(X)$ and $Q : B \to \Delta(Y)$,*

$$q^0 \succeq_{P \sqcup Q} r^1 \quad \Longleftrightarrow \quad F_q(\mu_{q,P}) \geq F_r(\mu_{r,Q}). \tag{BC}$$

*Boundary representatives are interpreted by Lemma 19.*

*Proof.* First suppose $q$ and $r$ have full support, allowing singleton action sets. At prior $q \otimes r$, quasi-additivity and zero normalisation give

$$F_{q\otimes r}(\mu_{q,P} \otimes \delta_r) = F_q(\mu_{q,P}), \qquad F_{q\otimes r}(\delta_q \otimes \mu_{r,Q}) = F_r(\mu_{r,Q}).$$

These values are in the same fibre $\mathcal{M}_{q\otimes r}$, so Lemma 15 gives

$$(q \otimes r)^0 \succeq_{(P\otimes U_B)\sqcup(U_A\otimes Q)} (q \otimes r)^1 \quad \Longleftrightarrow \quad F_q(\mu_{q,P}) \geq F_r(\mu_{r,Q}).$$

It remains to transfer the product-lifted comparison back to the original two blocks. For the $A$-side pair, Clause (DP-act) of Condition (P4) applied to the projection $\pi_A : A \times B \to A$ gives

$$(q \otimes r, P \otimes U_B) \succeq (q, P)$$

in block comparisons. Conversely, let $T : A \to \Delta(A \times B)$ be

$$T((a, b) \mid a') = \mathbf{1}_{\{a=a'\}} r(b).$$

Then $qT = q \otimes r$, and the Bayesian pushforward $T^q P$ has row $P(\cdot \mid a)$ at every $(a, b)$. Thus $T^q P$ is $P \otimes U_B$ after the canonical one-record identification of the $B$-experiment, and (DP-act) gives the reverse comparison. Equivalently, if the harmless record identification is not made, Lemma 12 identifies $T^q P$ and $P \otimes U_B$ because they induce the same posterior law at $q \otimes r$. Hence $(q \otimes r, P \otimes U_B)$ and $(q, P)$ are indifferent in every relevant block comparison. The same projection/embedding argument identifies $(q \otimes r, U_A \otimes Q)$ with $(r, Q)$.

Place the four pairs $(q, P)$, $(q \otimes r, P \otimes U_B)$, $(r, Q)$, and $(q \otimes r, U_A \otimes Q)$ as blocks of one finite block environment. Lemma 10 and transitivity transfer the two pairwise indifferences to the two-block comparison, proving (BC) for full-support priors.

For arbitrary priors, let $A_+ := \mathrm{supp}(q)$ and $B_+ := \mathrm{supp}(r)$. Lemma 19 identifies $(q, P)$ with $(q|_{A_+}, P|_{A_+})$, identifies $(r, Q)$ with $(r|_{B_+}, Q|_{B_+})$, and identifies the two values $F_q(\mu_{q,P})$ and $F_r(\mu_{r,Q})$ with their support-face values. Applying the full-support bridge on $A_+$ and $B_+$ and pulling back by Lemma 19 proves the claim. Singleton supports use the same zero-representative convention as Lemma 21. $\square$

**Stage 4: branch coefficients, cocycle, and support faces.** Insertion into a reached branch is affine. Its linear part is evaluated on finite atomic signed posterior laws; the spanning argument shows that the resulting positive coefficient depends only on the prior and reached posterior. Path independence gives the cocycle and the normalized chain rule. The subsequent cardinal correction aligns auxiliary chain scales on nested faces; it does not add a behavioural assumption or change the ordinal representative.

**Lemma 24** (Branch aggregation)**.** *Let $P_1 : A \to \Delta(X_1)$ be a first-stage channel at full-support $q$, with branch probabilities $m(x)$ and posteriors $r_x$. There are positive branch coefficients*

*$\beta(q, r_x)$, depending only on $q$ and the reached posterior $r_x$, such that for every continuation profile $\{Q^x\}_{x:m(x)>0}$,*

$$F_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = F_q(\mu_{q,P_1}) + \sum_{x:m(x)>0} m(x)\, \beta(q, r_x)\, F_{r_x}(\mu_{r_x,Q^x}).$$

*If $r_x$ has full support, $\beta(q, r_x)$ is the full-support tangent coefficient. If $1 < |\operatorname{supp}(r_x)| < |A|$, it is the full-to-face coefficient. If $|\operatorname{supp}(r_x)| = 1$, the value $F_{r_x}$ is identically zero and $\beta(q, r_x)$ is an arbitrary positive convention.*

*Proof.* Fix the first-stage channel $P_1$ and its branch structure $(m(x), r_x)_{x:m(x)>0}$. For any profile of continuation channels $\{Q^x\}$, the compound posterior law at prior $q$ is

$$\mu_{q,P_1 \triangleright \{Q^x\}} = \sum_{x:m(x)>0} m(x)\, \mu_{r_x,Q^x}.$$

*Setup: linear part of affine functional.* Since $F_q$ is affine on $\mathcal{M}_q$ (Lemma 15), it extends uniquely to an affine functional on the affine hull of $\mathcal{M}_q$. Write $L_q$ for the linear part: for any $\mu, \mu' \in \mathcal{M}_q$,

$$F_q(\mu) - F_q(\mu') = L_q(\mu - \mu').$$

*Branch-affine structure.* Fix a positive-probability branch $\bar{x}$ and fix continuation channels $H^x : A \to \Delta(Z^x)$ in all other branches $x \neq \bar{x}$. Let $r := r_{\bar{x}}$ and $m := m(\bar{x})$. If $\operatorname{supp}(r)$ is a singleton, then $\mathcal{M}_r = \{\delta_r\}$ and $F_r(\delta_r) = 0$ by the singleton zero-normalisation convention; such branches contribute zero and their coefficient convention is recorded in Case 1 below. For the branch-affine argument in this paragraph, assume $|\operatorname{supp}(r)| \geq 2$. If $r$ is a boundary posterior, $F_r$ and continuation comparisons at $r$ are understood through Lemma 19.

Let

$$\rho_H := \sum_{x \neq \bar{x}} m(x)\, \mu_{r_x,H^x}.$$

For any $\nu \in \mathcal{M}_r$, the aggregate posterior law produced by using branch law $\nu$ in branch $\bar{x}$ and the fixed continuations elsewhere is

$$T_H(\nu) := m\nu + \rho_H \in \mathcal{M}_q.$$

Define $g : \mathcal{M}_r \to \mathbb{R}$ by

$$g(\nu) := F_q(T_H(\nu)).$$

Thus $g$ depends only on $\nu$, and $g = F_q \circ T_H$ is continuous and affine.

We now show that $g$ and $F_r$ represent the same weak order on $\mathcal{M}_r$. For $\nu, \nu' \in \mathcal{M}_r$, choose finite continuation channels implementing them at prior $r$; when $r$ is boundary these are implemented on $B = \operatorname{supp}(r)$ and extended arbitrarily to $A$, as licensed by Lemma 19. Padding record alphabets if needed, Condition (P5) compares the two continuation profiles that differ only in branch $\bar{x}$. If $F_r(\nu) > F_r(\nu')$, Condition (P5) gives $g(\nu) > g(\nu')$. Conversely, if $g(\nu) \geq g(\nu')$ but $F_r(\nu) < F_r(\nu')$, applying the strict part of Condition (P5) to the swapped branch comparison gives $g(\nu') > g(\nu)$, a contradiction. Hence $g(\nu) \geq g(\nu')$ iff $F_r(\nu) \geq F_r(\nu')$.

Since $|\operatorname{supp}(r)| \geq 2$, local non-triviality on the support face and Lemmas 19 and 16 imply $F_r$ is nonconstant. The continuous affine functions $g$ and $F_r$ are therefore nonconstant affine representatives of the same weak order on $\mathcal{M}_r$. Herstein–Milnor uniqueness gives

$$g(\nu) = \alpha\, F_r(\nu) + \gamma$$

for some $\alpha > 0$ and $\gamma \in \mathbb{R}$.

*The coefficient $\alpha$ is independent of the fixed continuations.* Write $g^{(1)}$ and $g^{(2)}$ for the functionals arising from two different choices $\{H^x\}$ and $\{\widetilde{H}^x\}$ in the other branches. Both represent the same order as $F_r$, so $g^{(1)} = \alpha_1 F_r + \gamma_1$ and $g^{(2)} = \alpha_2 F_r + \gamma_2$ with $\alpha_1, \alpha_2 > 0$. To show $\alpha_1 = \alpha_2$, consider the difference $g^{(1)} - g^{(2)} = (\alpha_1 - \alpha_2) F_r + (\gamma_1 - \gamma_2)$. For any $\nu \in \mathcal{M}_r$,

$$g^{(1)}(\nu) - g^{(2)}(\nu) = F_q\Big(m(\bar{x})\nu + \textstyle\sum_{x \neq \bar{x}} m(x)\mu_{r_x, H^x}\Big) - F_q\Big(m(\bar{x})\nu + \textstyle\sum_{x \neq \bar{x}} m(x)\mu_{r_x, \widetilde{H}^x}\Big).$$

Both arguments are posterior laws in $\mathcal{M}_q$, differing only in the fixed non-$\bar{x}$ components. Since $L_q$ has already been defined on the affine-hull difference space,

$$g^{(1)}(\nu) - g^{(2)}(\nu) = L_q(\rho_1 - \rho_2),$$

where $\rho_i$ is the fixed non-$\bar{x}$ background law for $g^{(i)}$. This difference is independent of $\nu$. Since $F_r$ is nonconstant in the present nondegenerate case, $\alpha_1 = \alpha_2$.

*Path-independence of the branch coefficient via tangent-space decomposition.* We now prove that $\alpha / m(\bar{x})$ depends only on the pair $(q, r_{\bar{x}})$, not on the first-stage experiment. This is the key step that converts ordinal branchwise monotonicity into a cardinal aggregation formula.

*Tangent space at a posterior.* For a posterior $r \in \Delta(A)$ with support $B := \mathrm{supp}(r)$, define the *tangent space*:

$$V_r := \mathrm{span}\{\nu - \nu' : \nu, \nu' \in \mathcal{M}_r\}.$$

Elements of $V_r$ are signed measures with total mass zero and barycentre zero.

*Tangent-space identification.* For any $r$ with full support, we claim

$$V_r = W := \{\eta \in \mathrm{span}\, \Delta_f(\Delta(A)) : \eta(\Delta(A)) = 0,\ \textstyle\int p \, d\eta(p) = 0\},$$

the space of finite-support signed measures on $\Delta(A)$ with zero total mass and zero barycentre. Indeed, if $\nu, \nu' \in \mathcal{M}_r$, then $\nu - \nu'$ has total mass $1 - 1 = 0$ and barycentre $r - r = 0$, so $V_r \subset W$. Conversely, given any $\eta \in W$: if $\eta = 0$ then $\eta \in V_r$ trivially. Otherwise, since $\eta$ has finite support, write its Jordan decomposition $\eta = \eta^+ - \eta^-$ where $\eta^+, \eta^- \geq 0$ are finite-support positive measures with equal total mass $c > 0$. Normalise $\zeta^+ := \eta^+/c$ and $\zeta^- := \eta^-/c$; both are finite-support probability measures on $\Delta(A)$ with the same barycentre $b := \int p \, d\zeta^+(p) = \int p \, d\zeta^-(p)$. Since $b \in \Delta(A)$, the set $\{a : b(a) > 0\}$ is nonempty. Because $r$ has full support,

$$\theta := \min\Big\{1, \min_{a : b(a) > 0} \frac{r(a)}{b(a)}\Big\} > 0.$$

Choose $0 < \lambda < \theta$ and set

$$\tau := \frac{r - \lambda b}{1 - \lambda}.$$

Then $1 - \lambda > 0$, $\sum_a \tau(a) = 1$, and $\tau(a) \geq 0$ for every $a$: if $b(a) > 0$ this follows from $\lambda < r(a)/b(a)$, while if $b(a) = 0$ the numerator is $r(a) > 0$. Hence $\tau \in \Delta(A)$. Define

$$\nu := \lambda \zeta^+ + (1 - \lambda)\delta_\tau, \qquad \nu' := \lambda \zeta^- + (1 - \lambda)\delta_\tau.$$

Both are finite-support probability measures on $\Delta(A)$ with barycentre $\lambda b + (1 - \lambda)\tau = r$, so $\nu, \nu' \in \mathcal{M}_r$. Then $\nu - \nu' = \lambda(\zeta^+ - \zeta^-) = (\lambda/c)\eta$, hence $\eta = (c/\lambda)(\nu - \nu') \in V_r$. Thus $W \subset V_r$ and $V_r = W$. Since $W$ depends only on the full-support property (not on the specific prior), $V_q = V_r = V_s = W$ for all full-support priors.

*Case 1: Degenerate branch posterior ($|B| = 1$).* If $r = \delta_a$ for some action $a$, then $\mathcal{M}_{\delta_a} = \{\delta_{\delta_a}\}$ is a singleton: the only Bayes-plausible law with barycentre $\delta_a$ is the point mass at $\delta_a$ itself. Hence $V_r = \{0\}$, and every continuation experiment at prior $\delta_a$ gives the same posterior law $\delta_{\delta_a}$. We set $F_{\delta_a}(\delta_{\delta_a}) = 0$. For such branches, the term $m(x)\beta(q, r_x) F_{r_x}(\mu_{r_x, Q^x})$ is zero under any convention for $\beta(q, r_x)$.

For the aggregation formula below, assign any positive value to $\beta(q, \delta_a)$; the choice is not behaviourally identified because the corresponding continuation value is always zero.

*Case 2: Non-full-support branch posterior* $(1 < |B| < |A|)$*.* If $r$ has support $B \subsetneq A$ with $|B| \geq 2$, apply Lemma 19. Continuation comparisons at prior $r$ reduce to those at $r|_B \in \Delta(B)$, with representative $F_r := F^B_{r|_B} \circ \pi_B$.

For any finite action set $C$, write

$$W_C := \{\sigma \in \operatorname{span} \Delta_f(\Delta(C)) : \sigma(\Delta(C)) = 0,\ \textstyle\int_{\Delta(C)} p \, d\sigma(p) = 0\}.$$

Let $\jmath_B : \Delta(B) \to \Delta(A)$ be the face inclusion, extending each posterior by zero outside $B$, and let $i_B := (\jmath_B)_\#$ act on finite signed posterior laws. By Lemma 19(iv), $F^A_r = F^B_{r|_B} \circ \pi_B$, so on face directions

$$L^A_r(i_B\sigma) = L^B_{r|_B}(\sigma) \qquad (\sigma \in W_B).$$

Since $|B| \geq 2$ and $r|_B$ is full support on $B$, local non-triviality and Lemma 16 give

$$L^B_{r|_B}(\chi_{r|_B} - \delta_{r|_B}) > 0,$$

so $L^B_{r|_B} \neq 0$.

*Full-to-face scalar.* Fix any full-support $q \in \Delta(A)$. The boundary posterior $r$ is reachable from $q$: choose

$$0 < \varepsilon < \min\Big\{1, \min_{b \in B} \frac{q(b)}{r(b)}\Big\}$$

and set $P_1(1 \mid a) = \varepsilon r(a)/q(a)$, with $r(a) = 0$ for $a \notin B$. Then branch 1 has positive probability $\varepsilon$ and posterior $r$. For any nonzero $\sigma \in W_B$, the tangent-space identification on the face $B$ gives $\sigma = t(\nu - \nu')$ for some $t > 0$ and $\nu, \nu' \in \mathcal{M}^B_{r|_B}$. Implement $\nu, \nu'$ by continuation channels on $B$ and extend them arbitrarily to $A$. Lemma 19 identifies their comparison at the boundary prior $r$ with their comparison at $r|_B$.

Condition (P5) applied to the branch reaching $r$, with all other branches fixed, gives the positive sign direction for feasible differences. The negative direction follows by swapping the two continuation laws, and equality follows by applying the weak part of Condition (P5) in both directions. Extending by positive scalar multiples from feasible differences to all of $W_B$, $L^A_q \circ i_B$ and $L^B_{r|_B}$ have the same sign partition on $W_B$. Since $L^B_{r|_B} \neq 0$, both are nonzero and there is a unique $\beta_A(q, r) > 0$ such that

$$L^A_q(i_B\sigma) = \beta_A(q, r) L^B_{r|_B}(\sigma) \qquad (\sigma \in W_B).$$

This is the ambient full-to-face coefficient used for boundary branches.

*Case 3: Full-support branch posterior* $(B = A)$*.* This is the main case. Suppose $r$ has full support on $A$.

*Claim 1: aggregate change from branch variation.* If a first-stage branch reaches posterior $r$ with probability $m$, and the branch continuation law changes from $\nu$ to $\nu'$ (both in $\mathcal{M}_r$), then the aggregate posterior law changes by $m(\nu - \nu') \in m \cdot V_r$. This difference depends only on $(m, \nu, \nu', r)$—not on which experiment produced the branch, nor on the other branches.

*Proof of Claim 1.* The aggregate changes from $m\nu + \rho$ to $m\nu' + \rho$, where $\rho$ collects the other branches. The difference is $m(\nu - \nu')$.

*Claim 2:* $L_q|_{V_r}$ *is a positive scalar multiple of* $L_r|_{V_r}$*.*

*Proof of Claim 2.* We first show that every direction in $V_r$ is proportional to a feasible difference. By the Tangent-space identification above, any nonzero $\sigma \in V_r$ can be written as $\sigma = t(\nu - \nu')$ for some $t > 0$ and $\nu, \nu' \in \mathcal{M}_r$. Hence every nonzero $\sigma \in V_r$ is a positive scalar multiple of some feasible difference.

By Condition (P5) (branchwise monotonicity), if $\nu \succ_r \nu'$ (i.e. $F_r(\nu) > F_r(\nu')$, equivalently $L_r(\nu - \nu') > 0$), then the aggregate comparison ranks the $\nu$-continuation strictly higher. Since the aggregate value difference is $L_q(m(\nu - \nu')) = mL_q(\nu - \nu')$ with $m > 0$, we have $L_q(\nu - \nu') > 0$ whenever $L_r(\nu - \nu') > 0$.

*Reverse and equality directions.* If $L_r(\nu - \nu') < 0$, completeness of $\succeq_r$ gives $\nu' \succ_r \nu$, and Condition (P5) applied to the reversed comparison gives $L_q(\nu' - \nu) > 0$, i.e. $L_q(\nu - \nu') < 0$. If $L_r(\nu - \nu') = 0$, then $\nu \sim_r \nu'$, so both $\nu \succeq_r \nu'$ and $\nu' \succeq_r \nu$; applying Condition (P5) weakly in both directions gives $L_q(\nu - \nu') \geq 0$ and $L_q(\nu - \nu') \leq 0$, hence $L_q(\nu - \nu') = 0$.

Since every nonzero $\sigma \in V_r$ is proportional to some feasible difference, the sign agreement extends from feasible differences to all of $V_r$: for any $\sigma \in V_r$, $L_q(\sigma) > 0$ iff $L_r(\sigma) > 0$, $L_q(\sigma) < 0$ iff $L_r(\sigma) < 0$, and $L_q(\sigma) = 0$ iff $L_r(\sigma) = 0$.

Two nonzero linear functionals on a vector space that have the same positive half-space must be positive scalar multiples of each other. (If $L_r \neq 0$, then $\ker L_q = \ker L_r$ and they lie in the same half-space, so $L_q = cL_r$ for some $c > 0$.) Since $r$ has full support and local non-triviality gives $F_r(\chi_r) > F_r(\delta_r) = 0$, we have $L_r \neq 0$. Hence there exists $\beta(q, r) > 0$ such that

$$L_q(\sigma) = \beta(q, r)\, L_r(\sigma) \qquad \text{for all } \sigma \in V_r.$$

*Path-independence.* Consider any first-stage experiment reaching posterior $r$ in some branch with probability $m$. Changing the branch continuation from $\nu$ to $\nu'$ changes the aggregate value by

$$L_q(m(\nu - \nu')) = m\, \beta(q, r)\, L_r(\nu - \nu') = m\, \beta(q, r)\, (F_r(\nu) - F_r(\nu')).$$

The coefficient $m\, \beta(q, r)$ depends only on $(q, r, m)$. In particular, $\beta(q, r)$ is independent of which experiment reached $r$ and independent of the other branches.

Comparing with the earlier notation $g(\nu) = \alpha F_r(\nu) + \gamma$: the value difference is $g(\nu) - g(\nu') = \alpha(F_r(\nu) - F_r(\nu'))$, so $\alpha = m\, \beta(q, r)$ and hence $\alpha/m = \beta(q, r)$ universally. The coefficient $\beta(q, r_{\bar{x}})$ is defined as the scalar from $L_q|_{V_{r_{\bar{x}}}} = \beta(q, r_{\bar{x}}) L_{r_{\bar{x}}}|_{V_{r_{\bar{x}}}}$; it depends only on $q$ and $r_{\bar{x}}$, not on the experiment that produced the branch.

*Aggregation formula.* Let

$$\mu^0 := \sum_{x:m(x)>0} m(x)\delta_{r_x} = \mu_{q,P_1}.$$

For each positive-probability branch let $B_x := \mathrm{supp}(r_x)$. If $B_x = A$, set $v_x := \mu_{r_x,Q^x} - \delta_{r_x} \in W_A$. If $1 < |B_x| < |A|$, set

$$\bar{v}_x := \pi_{B_x}(\mu_{r_x,Q^x}) - \delta_{r_x|_{B_x}} \in W_{B_x}, \qquad v_x := i_{B_x}\bar{v}_x \in W_A.$$

If $|B_x| = 1$, set $v_x := 0$. Then

$$\mu_{q,P_1 \triangleright \{Q^x\}} = \mu^0 + \sum_{x:m(x)>0} m(x) v_x.$$

Since $V_q = W_A$, affinity of $F_q$ gives

$$F_q(\mu_{q,P_1 \triangleright \{Q^x\}}) - F_q(\mu^0) = \sum_{x:m(x)>0} m(x) L_q^A(v_x).$$

For a full-support branch $B_x = A$, Case 3 gives

$$L_q^A(v_x) = \beta(q, r_x) L_{r_x}^A(v_x) = \beta(q, r_x) F_{r_x}(\mu_{r_x,Q^x}).$$

For a nondegenerate boundary branch $1 < |B_x| < |A|$, Case 2 gives

$$L_q^A(v_x) = L_q^A(i_{B_x}\bar{v}_x) = \beta_A(q, r_x) L_{r_x|_{B_x}}^{B_x}(\bar{v}_x) = \beta_A(q, r_x) F_{r_x}(\mu_{r_x,Q^x}).$$

The last equalities use zero normalisation: Lemma 16 in full support, Lemma 19 plus Lemma 16 on the support face in the boundary case, and the singleton convention when $|B_x| = 1$. Therefore

$$F_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = F_q(\mu_{q,P_1}) + \sum_{x:m(x)>0} m(x)\, \beta(q, r_x)\, F_{r_x}(\mu_{r_x,Q^x}),$$

where $\beta(q, r_x)$ denotes the Case 3 coefficient when $B_x = A$, the Case 2 full-to-face coefficient when $1 < |B_x| < |A|$, and any positive singleton convention when $|B_x| = 1$. □

**Lemma 25** (Branch-coefficient cocycle and normalised chain rule)**.** *Let the full-support and full-to-face coefficients be those constructed in Lemma 24, applied in the relevant ambient simplex or intrinsically on the relevant support face. If $q$ has full support and $r, s$ are nondegenerate priors with*

$$\operatorname{supp}(s) \subseteq \operatorname{supp}(r) \subseteq \operatorname{supp}(q),$$

*then the coefficients satisfy the cocycle identity*

$$\beta(q, s) = \beta(q, r)\beta(r, s).$$

*Hence $\beta(q, r) = a_q/a_r$ for positive scales $a_q$, with singleton scales fixed by convention. The rescaled values*

$$\widehat{F}_q(\mu) := \frac{F_q(\mu)}{a_q}$$

*satisfy the exact chain rule: for every first-stage channel $P_1$ at full-support $q$ and every continuation profile $\{Q^x\}_{x:m(x)>0}$,*

$$\widehat{F}_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = \widehat{F}_q(\mu_{q,P_1}) \; + \sum_{x:m(x)>0} m(x)\, \widehat{F}_{r_x}(\mu_{r_x,Q^x}).$$

*Proof. Full-support cocycle.* First consider reachable full-support $q, r, s \in \operatorname{int} \Delta(A)$ with $|A| \geq 2$. For such priors, all three tangent spaces coincide. By the tangent-space identification in Lemma 24, $V_q = V_r = V_s = W$ for all full-support priors. The full-support coefficient identities constructed there share a common domain $W$:

$$L_q|_W = \beta(q, r)L_r|_W, \qquad L_r|_W = \beta(r, s)L_s|_W.$$

Substituting the second into the first:

$$L_q|_W = \beta(q, r)\beta(r, s)L_s|_W.$$

Comparing with $L_q|_W = \beta(q, s)L_s|_W$ and using $L_s \neq 0$:

$$\beta(q, s) = \beta(q, r)\beta(r, s).$$

*Reachability.* The above uses the full-support coefficient construction for the pairs $(q, r)$, $(r, s)$, and $(q, s)$. Each invocation requires the pair to be reachable: the second prior must be a possible posterior of some experiment at the first prior. For any full-support $q, r$, construct a two-record experiment $P_1$ with $P_1(1 \mid a) = \varepsilon r(a)/q(a)$ for $0 < \varepsilon < \min\{1, \min_a q(a)/r(a)\}$. Record 1 has probability $\varepsilon$ and posterior $r$; record 0 has posterior $(q - \varepsilon r)/(1 - \varepsilon)$. Hence any full-support $r$ is reachable from any full-support $q$, so the construction applies to all full-support pairs. The cocycle therefore holds for all full-support triples.

Fix a full-support basepoint $q_0$, set $a_{q_0} = 1$, and for full-support $q$ define

$$a_q := \frac{1}{\beta(q_0, q)}.$$

By the full-support cocycle, this is equivalently $a_q = \beta(q, q_0)$, and

$$\beta(q, r) = \frac{\beta(q, q_0)}{\beta(r, q_0)} = \frac{a_q}{a_r}.$$

*Boundary coefficients and face scales.* If $r$ is nondegenerate with support $B \subsetneq A$, Lemma 24 constructs the ambient full-to-face coefficient $\beta_A(q, r) > 0$ by

$$L_q^A i_B = \beta_A(q, r) L^B_{r|_B}.$$

When the ambient action set is clear, write this coefficient as $\beta(q, r)$.

When nondegenerate boundary priors $r, s$ have common support $B$, $\beta_A(r, s)$ means the unique $c > 0$ satisfying

$$L_r^A(i_B\sigma) = c\, L_s^A(i_B\sigma) \qquad (\sigma \in W_B).$$

This is a scalar on the common face, not on all of $W_A$. Since

$$L_r^A \circ i_B = L^B_{r|_B}, \qquad L_s^A \circ i_B = L^B_{s|_B},$$

and the intrinsic face scalar satisfies $L^B_{r|_B}(\sigma) = \beta_B(r|_B, s|_B) L^B_{s|_B}(\sigma)$, we have

$$\beta_A(r, s) = \beta_B(r|_B, s|_B).$$

Fix the same full-support basepoint $q_0$ on $\Delta(A)$. If $r_0$ is any nondegenerate face basepoint with support $B$, then for every nondegenerate $r$ with support $B$,

$$\beta_A(q_0, r) = \beta_A(q_0, r_0)\, \beta_B(r_0|_B, r|_B).$$

Indeed, both sides are the coefficient multiplying $L^B_{r|_B}$ to obtain $L^A_{q_0} \circ i_B$, and $L^B_{r|_B} \neq 0$. Define the ambient boundary scale by

$$a_r := \frac{1}{\beta_A(q_0, r)}.$$

If the face scale is based at $r_0$ with $a_{r_0} := 1/\beta_A(q_0, r_0)$, then

$$a^B_{r|_B} := \frac{a_{r_0}}{\beta_B(r_0|_B, r|_B)} = \frac{1}{\beta_A(q_0, r_0)\beta_B(r_0|_B, r|_B)} = \frac{1}{\beta_A(q_0, r)} = a_r.$$

Thus the ambient and intrinsic scales agree: treating $r$ as a boundary posterior in $\Delta(A)$ or as a full-support prior on $\Delta(B)$ yields the same normalised representative $\widehat{F}_r$.

*Extension to boundary posteriors.* If $r$ is nondegenerate with support $B \subsetneq A$, Lemma 24 defines $\beta(q, r) = \beta_A(q, r)$ by

$$L_q^A i_B = \beta_A(q, r) L^B_{r|_B}.$$

Using the full-support identity $L_q^A = \beta(q, q_0) L^A_{q_0}$ and restricting to $i_B(W_B)$,

$$L_q^A i_B = \beta(q, q_0) L^A_{q_0} i_B = \beta(q, q_0)\beta_A(q_0, r) L^B_{r|_B}.$$

Comparing with the definition of $\beta_A(q, r)$ and using $L^B_{r|_B} \neq 0$ gives

$$\beta(q, r) = \beta(q, q_0)\beta(q_0, r) = \frac{a_q}{a_r},$$

where $a_r = 1/\beta(q_0, r)$ by the boundary-scale definition above.

*Singleton scale conventions.* If $r = \delta_a$ is a singleton prior, set $a_{\delta_a} := a_{q_0} = 1$ and $\widehat{F}_{\delta_a} \equiv 0$. Define $\beta(q, \delta_a) := a_q / a_{\delta_a}$ as a convention. No behavioural branch coefficient is identified at a singleton target, but singleton terms in the aggregation and chain formulas are always zero.

*Mixed-support cocycle.* Let $r, s$ be nondegenerate with nested supports $C := \mathrm{supp}(s) \subseteq B := \mathrm{supp}(r)$, and write $s|_B$ for $s$ viewed as a boundary prior on the action set $B$. Interpret

$$\beta(r, s) := \beta_B(r|_B, s|_B),$$

the intrinsic full-to-face coefficient on action set $B$. Let $i_C^A : W_C \to W_A$ and $i_C^B : W_C \to W_B$ be the face inclusions. For every $\sigma \in W_C$,

$$L_q^A i_C^A \sigma = \beta(q, r) L^B_{r|_B} i_C^B \sigma = \beta(q, r)\beta(r, s) L^C_{s|_C} \sigma.$$

The direct full-to-face definition from Lemma 24 for the pair $(q, s)$ gives

$$L_q^A i_C^A \sigma = \beta(q, s) L_{s|_C}^C \sigma.$$

Since $L_{s|_C}^C \neq 0$, $\beta(q, s) = \beta(q, r)\beta(r, s)$. If $\operatorname{supp}(s) \not\subseteq \operatorname{supp}(r)$, then $\beta(r, s)$ is not behaviourally defined; singleton targets are also excluded from the behavioural cocycle because their scales are conventions only.

*Chain rule.* Applying Lemma 24 and substituting $\beta(q, r) = a_q/a_r$ gives

$$F_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = F_q(\mu_{q,P_1}) + \sum_{x:m(x)>0} m(x) \frac{a_q}{a_{r_x}} F_{r_x}(\mu_{r_x,Q^x}).$$

Dividing by $a_q$ and writing $\widehat{F}_q(\mu) := F_q(\mu)/a_q$ yields

$$\widehat{F}_q(\mu_{q,P_1 \triangleright \{Q^x\}}) = \widehat{F}_q(\mu_{q,P_1}) + \sum_{x:m(x)>0} m(x)\, \widehat{F}_{r_x}(\mu_{r_x,Q^x}). \qquad \square$$

**Lemma 26** (Coherent relabelling and face scales)**.** *The scales produced by Lemma 25 can be chosen simultaneously across all finite action sets so that:*

1. *for every bijection $\pi : A \to A'$,*
$$a_{q\pi}^{A'} = a_q^A;$$
2. *if $\iota : B \hookrightarrow A$ is an injection with $|B| \geq 2$, $r$ has full support on $B$, and $q$ has full support on $A$, then the full-to-face branch coefficient for reaching the face $\iota(B)$ is*
$$\beta_\iota(q, r) = \frac{a_q^A}{a_r^B}. \tag{FS}$$

*Singleton target scales remain a convention because every continuation value on a singleton fibre is zero.*

*Proof.* For every nondegenerate finite action set $A$, first choose the full-support scales supplied by the cocycle part of Lemma 25, so that

$$\beta_A(q, q') = \frac{a_q^A}{a_{q'}^A} \qquad (q, q' \in \operatorname{int} \Delta(A)).$$

They are unique up to multiplication by one positive constant depending on $A$. Normalise initially by $a_{u_A}^A = 1$, where $u_A$ is uniform. Exact relabelling invariance of the representatives implies relabelling invariance of the uniquely determined branch coefficients. Since bijections carry uniform priors to uniform priors, the initial scales are exactly invariant under bijections.

Let $\iota : B \hookrightarrow A$ be an injection between nondegenerate finite sets. By relabelling $B$ onto its image $\iota(B)$ and using Corollary 22, it is enough to define the coefficient for the inclusion of the face $\iota(B)$; the notation below keeps the original symbol $\iota$. Let $i_\iota$ denote the induced map on signed posterior-law tangent spaces. For full-support $q \in \Delta(A)$ and $r \in \Delta(B)$, let $\beta_\iota(q, r) > 0$ be the full-to-face scalar characterised by

$$L_q^A \circ i_\iota = \beta_\iota(q, r) L_r^B \quad \text{on } W_B.$$

The number

$$K(\iota) := \beta_\iota(q, r) \frac{a_r^B}{a_q^A} \tag{K}$$

is independent of both $q$ and $r$. Independence of $q$ follows by comparing two full-support ambient priors and using $L_q^A = (a_q^A/a_{q'}^A) L_{q'}^A$ on $W_A$. Independence of $r$ follows from same-face compatibility: if $r, s$ have full support on $B$, then $L_r^B = (a_r^B/a_s^B)L_s^B$, and hence $\beta_\iota(q, s) = \beta_\iota(q, r)a_r^B/a_s^B$.

The defects multiply under composition. If $C \overset{\jmath}{\hookrightarrow} B \overset{\iota}{\hookrightarrow} A$ and all three sets in the comparison are nondegenerate, then restricting first to $B$ and then to $C$ gives, for full-support $q, r, s$ on $A, B, C$ respectively,

$$\beta_{\iota\circ\jmath}(q, s) = \beta_\iota(q, r)\beta_\jmath(r, s).$$

Substituting the definition of $K$ and cancelling the intermediate scale $a_r^B$ gives

$$K(\iota \circ \jmath) = K(\iota)K(\jmath). \tag{KC}$$

Exact relabelling invariance shows that $K(\iota)$ depends only on $n := |A|$ and $m := |B|$; write it as $K_{n,m}$. Bijections give $K_{n,n} = 1$, and

$$K_{n,\ell} = K_{n,m}K_{m,\ell} \qquad (2 \le \ell \le m \le n).$$

Put $t_n := K_{n,2}$, with $t_2 = 1$. Then $K_{n,m} = t_n/t_m$. Finally replace every scale on an $n$-point action set by

$$\widetilde{a}_q^A := t_n a_q^A.$$

Under this change, the defect in (K) becomes

$$\widetilde{K}_{n,m} = K_{n,m}\frac{t_m}{t_n} = 1.$$

The rescaling is common to all priors on a fixed action set, so it leaves all within-set cocycle ratios unchanged; because $t_n$ depends only on cardinality, it also preserves exact relabelling invariance. Dropping tildes gives (FS). Singleton target scales may be set arbitrarily because all singleton continuation terms are zero. □

**Stage 5: compatibility collapse and entropy identification.** The product and sequential evaluations of full revelation must agree. Their compatibility produces a positive grouping-weight equation; evaluating one distribution by two groupings forces the product interaction to vanish and the chain scale to be universal. Entropy reduction and Faddeev's recursion then identify the remaining functional.

**Lemma 27** (Interaction collapse and universal chain scale)**.** *The universal interaction coefficient in* (QA) *is zero. Moreover, there exists $a > 0$ such that the coherently chosen chain scales satisfy $a_q = a$ for every nondegenerate full-support prior on every finite action set. The same value may be assigned to singleton priors. Consequently,*

$$F_{p\otimes r}(\mu \otimes \nu) = F_p(\mu) + F_r(\nu) \tag{PA}$$

*and the branch formula has a universal chain scale.*

*Proof.* Define, for every finite prior $q$ read on its support,

$$H(q) := F_q(\chi_q), \qquad Z(q) := 1 + \kappa H(q).$$

For a nondegenerate prior, local non-triviality gives $H(q) > 0$, and (POS) gives

$$Z(q) > 0. \tag{Z+}$$

For a singleton, $H(q) = 0$ and $Z(q) = 1$.

*Step 1: product revelation links the chain scale to $Z$.* Let $q \in \Delta(A)$ and $r \in \Delta(B)$ be nondegenerate and full support. Quasi-additivity applied to full revelation gives

$$H(q \otimes r) = H(q) + H(r) + \kappa H(q)H(r). \tag{QH}$$

Compute the same value sequentially. First reveal the $A$-coordinate and then reveal the $B$-coordinate in every branch. The first-stage value is $F_{q\otimes r}(\chi_q \otimes \delta_r) = H(q)$ by (QA). After observing $a$, the posterior is $e_a \otimes r$ on the face $\{a\} \times B$; support restriction and relabelling identify its continuation full-revelation value with $H(r)$. For the injection

$$\iota_a : B \hookrightarrow A \times B, \qquad b \mapsto (a, b),$$

Lemma 26 gives the branch coefficient

$$\beta_{\iota_a}(q \otimes r, r) = \frac{a^{A\times B}_{q\otimes r}}{a^B_r}.$$

The branch weights are $q(a)$ and the same coefficient and continuation value appear in every branch, so $\sum_a q(a) = 1$. Lemma 24 therefore yields

$$H(q \otimes r) = H(q) + \frac{a_{q\otimes r}}{a_r} H(r). \tag{SA}$$

Comparing (QH) and (SA), and using $H(r) > 0$,

$$\frac{a_{q\otimes r}}{a_r} = Z(q). \tag{R1}$$

Revealing $B$ first gives symmetrically

$$\frac{a_{q\otimes r}}{a_q} = Z(r). \tag{R2}$$

Hence $a_r Z(q) = a_q Z(r)$, so there is a universal constant $C > 0$ such that

$$a_q = C Z(q) \tag{AS}$$

for every nondegenerate full-support prior on every finite action set. Assign $a_q := C$ at singleton priors, where $Z(q) = 1$ and all continuation values are zero.

If $\kappa = 0$, then $Z \equiv 1$, equation (AS) already gives universal scales, and (QA) is exact product additivity. For the remaining argument suppose $\kappa \neq 0$.

*Step 2: the induced weight equation.* Let $q$ be full support on $A$, let $\mathcal{P} = \{A_1, \dots, A_m\}$ be a partition into nonempty cells, and put

$$p_i := q(A_i), \qquad q^i := q(\cdot \mid A_i), \qquad p := (p_1, \dots, p_m).$$

Let $G_{\mathcal{P}}$ reveal the block. Undetectable-distinctions neutrality gives $(q, G_{\mathcal{P}}) \sim (p, \mathrm{Id}_{\{1,\dots,m\}})$. The early cross-prior representation Lemma 23, importantly for the *unscaled* $F$'s, therefore gives

$$F_q(\mu_{q,G_{\mathcal{P}}}) = H(p). \tag{BG}$$

Append full revelation in every branch. The compound posterior law is $\chi_q$, so the chain formula and (AS) give

$$H(q) = H(p) + Z(q) \sum_{i=1}^{m} p_i \frac{H(q^i)}{Z(q^i)}. \tag{GR}$$

Define

$$w(q) := \frac{1}{Z(q)} > 0.$$

Using $\kappa H(x) = Z(x) - 1$ in (GR) gives

$$\frac{Z(p)}{Z(q)} = \sum_{i=1}^{m} p_i \frac{1}{Z(q^i)}.$$

Equivalently,

$$w(q) = w(p) \sum_{i=1}^{m} p_i w(q^i). \tag{W}$$

No continuity assertion is needed here.

*Step 3: a two-grouping argument forces $w \equiv 1$.* All distributions in this paragraph are read on their positive-probability supports, so the weight equation applies after support restriction. First, (W) applied to a product distribution gives

$$w(u \otimes v) = w(u)w(v), \tag{M}$$

because the first-coordinate block probabilities are $u$ and every conditional distribution is $v$.

Take arbitrary finite probability vectors $u$ and $v$, possibly on different alphabets, and write

$$x := w(u), \qquad y := w(v), \qquad p := (1/2, 1/2).$$

On the labelled disjoint union $(U \times U)^0 \sqcup (V \times V)^1$, form the distribution

$$T := \tfrac{1}{2}(u \otimes u)^0 \ \sqcup \ \tfrac{1}{2}(v \otimes v)^1.$$

Grouping first by the top block and using (M),

$$w(T) = w(p)\frac{x^2 + y^2}{2}. \tag{E1}$$

Now group first by the pair consisting of the top-block label and the first within-block coordinate. Its marginal is

$$S := \tfrac{1}{2}u^0 \ \sqcup \ \tfrac{1}{2}v^1,$$

and (W) gives

$$w(S) = w(p)\frac{x + y}{2}.$$

Conditional on a point in the first part of $S$, the remaining coordinate has distribution $u$; conditional on a point in the second part, it has distribution $v$. A second application of (W) therefore gives

$$w(T) = w(S)\frac{x + y}{2} = w(p)\left(\frac{x + y}{2}\right)^2. \tag{E2}$$

Since $w(p) > 0$, comparison of (E1) and (E2) yields

$$\frac{x^2 + y^2}{2} = \left(\frac{x + y}{2}\right)^2,$$

so $(x - y)^2 = 0$. As $u$ and $v$ were arbitrary, $w$ is constant on all finite probability vectors. At a singleton prior, $H = 0$, hence $w = 1$. Therefore $w(q) \equiv 1$.

Thus $Z(q) = 1$ and $\kappa H(q) = 0$ for every $q$. Local non-triviality gives a nondegenerate $q$ with $H(q) > 0$, hence $\kappa = 0$, contradicting the temporary alternative unless the interaction coefficient is zero. Equation (AS) now gives $a_q = C$ universally. Finally (QA) reduces to exact product additivity (PA). $\square$

**Lemma 28** (Entropy reduction and Faddeev identification)**.** *By Lemma 27, $a_q \equiv a$ is universal. Define*

$$\widehat{F}_q := F_q/a, \qquad H(q) := \widehat{F}_q(\chi_q),$$

*the rescaled value of full revelation. Then, for every channel $P$,*

$$\widehat{F}_q(\mu_{q,P}) = H(q) - \sum_{x : m_{q,P}(x) > 0} m_{q,P}(x)\, H(r_x^{q,P}). \tag{ER}$$

*Moreover $H$ satisfies Faddeev's recursion, so*

$$H(q) = \alpha H(q)$$

*for some $\alpha > 0$. Consequently,*

$$\widehat{F}_q(\mu_{q,P}) = \alpha I_{q,P}(A; X).$$

*Proof. Entropy-reduction formula.* Boundary values of $H$ are read through support restriction: if $q$ has support $B$, then $H(q) := H(q|_B)$, and $H(\delta_a) = 0$. Let $P : A \to \Delta(X)$ be any channel at full-support prior $q$. Append full revelation $\mathrm{Id}_A : A \to \Delta(A)$ after $P$: the compound $P \triangleright \{\mathrm{Id}_A\}_x$ (using $\mathrm{Id}_A$ in every branch) has $\mu_{r_x,\mathrm{Id}_A} = \chi_{r_x}$ and the overall posterior law is full revelation $\chi_q$. Applying the chain rule (Lemma 25) gives

$$\widehat{F}_q(\chi_q) = \widehat{F}_q(\mu_{q,P}) + \sum_{x:m(x)>0} m(x)\, \widehat{F}_{r_x}(\chi_{r_x}).$$

Hence

$$\widehat{F}_q(\mu_{q,P}) = H(q) - \sum_{x:m(x)>0} m(x)\, H(r_x).$$

If $q$ is on the boundary, restrict $q$ and $P$ to $B = \mathrm{supp}(q)$, apply the full-support formula on $\Delta(B)$, and pull the value and comparison back by Lemma 19; zero-probability rows and records do not affect either side of the displayed identity.

*Scaled cross-prior bridge.* Lemma 23 was proved for the unscaled values $F_q$. Since Lemma 27 has made $a_q \equiv a$ universal, it is equivalently valid for

$$V(q, P) := \widehat{F}_q(\mu_{q,P}) = F_q(\mu_{q,P})/a :$$

for all finite priors and channels,

$$q^0 \succeq_{P \sqcup Q} r^1 \quad \Longleftrightarrow \quad V(q, P) \geq V(r, Q).$$

*Faddeev recursion.* First let $q$ be full support and let $\mathcal{P} = \{A_1, \ldots, A_m\}$ be a partition into nonempty cells. Then $p_i := q(A_i) > 0$ for every $i$. Let $G_{\mathcal{P}} : A \to \Delta(\{1, \ldots, m\})$ reveal only the block index: $G_{\mathcal{P}}(i \mid a) = \mathbf{1}_{a \in A_i}$. Write

$$p_i := q(A_i), \qquad q^i := q(\cdot \mid A_i),$$

and let $p = (p_1, \ldots, p_m) \in \Delta(\{1, \ldots, m\})$. Consider the deterministic kernel $S : A \to \Delta(\{1, \ldots, m\})$ given by $S(i \mid a) = \mathbf{1}_{a \in A_i}$, so $qS = p$.

By Lemma 18, since $G_{\mathcal{P}}$ is $S$-measurable,

$$(q, G_{\mathcal{P}}) \sim (p, \mathrm{Id}_{\{1,\ldots,m\}})$$

in the two-block comparison. Applying the scaled form of Lemma 23 to the forward comparison gives $V(q, G_{\mathcal{P}}) \geq V(p, \mathrm{Id}_{\{1,\ldots,m\}})$. Applying the reverse-block form from Lemma 10 and then the same bridge to the reverse comparison gives $V(p, \mathrm{Id}_{\{1,\ldots,m\}}) \geq V(q, G_{\mathcal{P}})$. Therefore

$$\widehat{F}_q(\mu_{q,G_{\mathcal{P}}}) = \widehat{F}_p(\chi_p) = H(p).$$

The entropy-reduction formula applied to $G_{\mathcal{P}}$ gives

$$H(q) = \widehat{F}_q(\chi_q) = \widehat{F}_q(\mu_{q,G_{\mathcal{P}}}) + \sum_{i=1}^{m} p_i \widehat{F}_{q^i}(\chi_{q^i}) = H(p) + \sum_{i=1}^{m} p_i H(q^i).$$

This is Faddeev's recursion. For boundary $q$, restrict to $\mathrm{supp}(q)$, apply the same full-support recursion to the nonempty positive-mass intersections, and pull back by Lemma 19; equivalently, the recursion is read over positive-mass blocks only.

We now verify that $H$ is continuous on each $\Delta_n := \{q \in \Delta(A) : |A| = n\}$.

*Binary continuity via erasure calibrators.* Define the binary entropy function $h : [0,1] \to \mathbb{R}$ by $h(t) := H(t, 1-t)$. We show $h$ is continuous using prior-independent calibrators. By Lemma 16, support restriction, and $a > 0$, $H(q) \geq 0$ for every $q$, so $h(t) \geq 0$ on $[0,1]$. If $c < 0$, then

$$\{t : h(t) \geq c\} = [0,1], \qquad \{t : h(t) \leq c\} = \varnothing,$$

both closed. Now fix $c \geq 0$. By local non-triviality, there exists a binary prior $\theta = (s, 1-s)$ with $u := H(\theta) > 0$. Choose $n \in \mathbb{N}$ and $\lambda \in [0,1]$ such that $c = \lambda n u$ (take $n$ large enough that $c \leq nu$, then set $\lambda := c/(nu)$).

Let $w_n := \theta^{\otimes n}$ be the $n$-fold product prior on $B_n := \{0,1\}^n$. By product additivity, $H(w_n) = nH(\theta) = nu$. Define the erasure experiment $E_{n,\lambda} : B_n \to \Delta(\{*, 1, \ldots, 2^n\})$ which reveals the state fully with probability $\lambda$ and reveals nothing with probability $1-\lambda$. By the entropy-reduction formula,

$$\widehat{F}_{w_n}(\mu_{w_n, E_{n,\lambda}}) = \lambda H(w_n) = \lambda n u = c.$$

Now consider the block action space $A_c := \{0,1\} \sqcup B_n$ and the block-diagonal channel $K_c := \mathrm{Id}_{\{0,1\}} \sqcup E_{n,\lambda}$. For $t \in [0,1]$, let $q_t^0$ be the prior $(t, 1-t)$ supported on block $\{0,1\}$, and let $w_n^1$ be $w_n$ supported on block $B_n$. By the scaled form of Lemma 23 (which extends to boundary priors via Lemma 19),

$$q_t^0 \succeq_{K_c} w_n^1 \quad \iff \quad V(q_t, \mathrm{Id}_{\{0,1\}}) = h(t) \geq c = V(w_n, E_{n,\lambda}).$$

Since $K_c$ is fixed and the map $t \mapsto q_t^0$ is continuous in $\ell_1$, Condition (P2) implies that the set $\{t \in [0,1] : h(t) \geq c\}$ is closed. Similarly, $\{t \in [0,1] : h(t) \leq c\}$ is closed. Thus all real superlevel and sublevel sets of $h$ are closed, so $h$ is both upper and lower semicontinuous on $[0,1]$, hence continuous.

*Continuity on higher simplices by induction.* For $n \geq 3$, write any $q \in \Delta_n$ as $q = (q_1, (1-q_1)\bar{q})$ where $\bar{q} := (q_2/(1-q_1), \ldots, q_n/(1-q_1)) \in \Delta_{n-1}$ for $q_1 < 1$. The Faddeev recursion with the binary partition $\{\{1\}, \{2, \ldots, n\}\}$ gives

$$H(q) = h(q_1) + (1-q_1)H(\bar{q}).$$

On the region $0 < q_1 < 1$, this is continuous: $h$ is continuous by binary continuity, $H|_{\Delta_{n-1}}$ is continuous by the induction hypothesis, and $q \mapsto (q_1, \bar{q})$ is continuous. If $q_1 = 0$, support restriction, equivalently the positive-mass-block convention in the recursion, gives $H(q) = H(\bar{q})$ and $h(0) = 0$, so continuity at that face follows from the induction hypothesis. At $q_1 = 1$, the induction hypothesis and compactness of $\Delta_{n-1}$ imply $H$ is bounded on $\Delta_{n-1}$, so $(1-q_1)H(\bar{q}) \to 0$ as $q_1 \to 1$. Thus $H(q) \to h(1) = 0 = H(\delta_1)$. The other boundary faces are handled by support restriction, relabelling on the support, and pulling back. This completes the induction.

We have verified the hypotheses of the strong-additivity form of Faddeev's finite entropy characterisation, as stated for example by Baez–Fritz–Leinster [1, Theorem 6], following Faddeev's original theorem [2]: a function on finite probability vectors that is continuous on each closed simplex, invariant under bijections, expansive under adjoining zero-probability states, zero on point masses, and strongly additive is a nonnegative multiple of Shannon entropy. Here invariance under bijections follows from Corollary 22, with boundary cases obtained by restricting to the support and pulling back; expansibility follows from support restriction; and the grouping recursion above is precisely strong additivity, read over positive-mass blocks. Faddeev's theorem therefore gives $H(q) = \alpha H(q)$ for some $\alpha \geq 0$. Local non-triviality (Condition (P1)) forces $\alpha > 0$.

*Mutual information.* Substituting $H = \alpha H$ into the entropy-reduction formula yields

$$\widehat{F}_q(\mu_{q,P}) = \alpha H(q) - \sum_{x:m_{q,P}(x)>0} m_{q,P}(x)\alpha H(r_x^{q,P}) = \alpha(H(q) - \mathbb{E}[H(r_X)]) = \alpha I_{q,P}(A;X). \quad \square$$

**Lemma 29** (Fixed-environment mutual-information representation)**.** *For every finite channel $P : A \to \Delta(X)$ and all $q, q' \in \Delta(A)$,*

$$q \succeq_P q' \quad \Longleftrightarrow \quad I_{q,P}(A;X) \geq I_{q',P}(A;X).$$

*Proof.* Let

$$V(s,R) := \widehat{F}_s(\mu_{s,R}),$$

with boundary priors interpreted by Lemma 19. Lemma 23, divided by the universal scale from Lemma 27, gives the cross-prior block-comparison bridge: for finite action sets, priors $s, t$, and channels $R, S$,

$$s^0 \succeq_{R \sqcup S} t^1 \quad \Longleftrightarrow \quad V(s,R) \geq V(t,S),$$

including boundary priors by support restriction. Lemma 28 gives, for every finite prior/channel pair,

$$V(s,R) = \alpha I_{s,R}$$

with $\alpha > 0$.

Fix $P : A \to \Delta(X)$ and $q, q' \in \Delta(A)$. By the duplication clause of Condition (P3),

$$q \succeq_P q' \quad \Longleftrightarrow \quad q^0 \succeq_{P \sqcup P} (q')^1.$$

Applying this bridge with $(s,R) = (q,P)$ and $(t,S) = (q',P)$ gives

$$q^0 \succeq_{P \sqcup P} (q')^1 \quad \Longleftrightarrow \quad V(q,P) \geq V(q',P).$$

Substituting $V = \alpha I$ and using $\alpha > 0$ yields

$$q \succeq_P q' \quad \Longleftrightarrow \quad I_{q,P}(A;X) \geq I_{q',P}(A;X).$$

If $q$ or $q'$ is on the boundary, the bridge is read after restricting each pair to its positive-probability support and then pulling the comparison back by Lemma 19. Deleting zero-probability actions does not change mutual information, and the Condition (P3) equivalence remains the ambient fixed-channel comparison for $P$. $\square$

*Proof of Proposition 9.* That (ii) implies (i) is the necessity verification at the start of this appendix. That (i) implies (ii) is Lemma 29. For the moreover clause, apply (ii) to the fixed block environment $\bigsqcup_{k \in K} P_k$: a lottery supported on block $i$ has constant block label, so

$$I_{q_i^i, \bigsqcup_{k \in K} P_k} = I_{q_i, P_i}(A_i; X_i),$$

and similarly for block $j$. $\square$

## References


[1] John C. Baez, Tobias Fritz, and Tom Leinster. A characterization of entropy in terms of information loss. *Entropy*, 13(11):1945–1957, 2011.

[2] D. K. Faddeev. On the concept of entropy of a finite probabilistic scheme. *Uspekhi Matematicheskikh Nauk*, 11(1):227–231, 1956. In Russian.

[3] I. N. Herstein and John Milnor. An axiomatic approach to measurable utility. *Econometrica*, 21(2):291–297, 1953.